Neural means and kernel corrections for operator learning

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Abstract

We combine neural network means with exact Matérn kernel regressions of their residuals and of their learned features, and evaluate the pairing on two public emulation problems with published baselines: the structural-mechanics benchmark of de Hoop et al. and the OCO-2 radiative-transfer emulator of Lamminpää et al. On structural mechanics the pipeline reaches 4.572% test error with a standard deviation of 0.010 points over ten independently seeded replicates: below every published method except the best one, from which it is not separable once the test block's own sampling error is counted. It reaches $5.433 \pm 0.093\%$, again over ten seeds, against a published 6.49% in the low-data regime. On OCO-2, replicated at ten seeds on each of the three spectral bands, it improves on the published Gaussian-process emulator on that problem's own test points and on both of its error metrics at once — on every seed of two bands and on nine of ten of the third; the same kernel that trails the network tenfold on the raw state overtakes it on the network's features, and we measure why (the target's squared native-space norm drops about fortyfold at fixed effective dimension) and prove the mechanism. Where the two families tie instead, the residuals of every architecture we train correlate above 0.83 at every seed, their shared component is flat in diversity and sample size, and the hardest cases are hard for every member at once, which reads the published plateau as a property of the data. The seed spread of our own pipeline is some sixty times smaller than the span of the published architectures, so that reading does not rest on run-to-run noise. The supporting results cover the deployed estimators themselves: a second-moment identity whose stacking prediction we check prospectively at every seed, where it tracks the realized risk to within a dispersion factor of 0.938 with a seed spread of 0.003; finite-sample guarantees with measured constants for the fitted stacks, the per-coordinate selection and the grid-tuned correction; an optimal-recovery certificate; and a distribution-free coverage band whose hypotheses the calibration protocol satisfies exactly, which is the only uncertainty signal that survives our tests and whose observed coverage falls inside the exact finite-sample band at ten seeds out of ten. Every stated result is also checked numerically.

1 Introduction

Operator learning constructs fast surrogates for the solution maps of parametric partial differential equations and physical forward models. Neural operators [11, 15, 16] learn their own representations; kernel and Gaussian-process methods [3, 20, 21] come with exact solves and error certificates and a strong record on the same benchmarks. This paper combines the two: a neural network mean, trained in the metric the application reports, with an exact Matérn kernel

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Table 1: The two problems studied in depth (OCO-2, structural mechanics) and the rest of the operator-learning suite, with the best published mean relative L^2 test error and ours. OCO-2 rows compare against the Gaussian-process emulator of Lamminpää et al. [13] on its own test points, in that paper's reduced-coefficient metric; Section 5 reports both of its metrics, on which one surrogate wins both on each of the three bands. The published suite numbers are per-problem methods tuned by their authors; ours outside the two problems studied in depth come from one recipe run unchanged (a Matérn kernel at the median length scale, fit capped at 6000 points, a residual MLP mean, outputs reduced by PCA, the stage selected on validation). Burgers and Darcy use slightly smaller training splits than their baselines (800 and 824 pairs) and Darcy a different grid, from the copies we could obtain.

Problem	Map	Best published	This work
OCO-2, O2 band	atmospheric state $\to$ radiance	16.89%	$4.12 \pm 0.05\%$
OCO-2, weak CO_2	atmospheric state $\to$ radiance	24.06%	$16.28 \pm 0.06\%$
OCO-2, strong CO_2	atmospheric state $\to$ radiance	16.14%	$8.08 \pm 0.01\%$
Structural mechanics	boundary load $\to$ stress	4.55% (PARA-Net)	$4.572 \pm 0.010\%$
Burgers	initial condition $\to$ solution	1.93% (FNO)	2.38%
Darcy flow	permeability $\to$ pressure	2.32% (POD-DeepONet)	2.01%
Advection I	smooth pulse $\to$ solution	$\approx 0\%$ (kernel)	—
Advection II	discontinuous pulse $\to$ solution	11.28% (linear kernel)	11.29%
Helmholtz	wavespeed $\to$ field	1.00% (kernel)	1.75%
Navier–Stokes	vorticity $\to$ vorticity	0.12% (kernel)	0.51%

regression of its residual and of its learned features. We evaluate the combination in depth on two public problems with published baselines — the structural-mechanics benchmark of de Hoop et al. [8] and the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [13] — and, with one unchanged recipe, across the seven-benchmark operator-learning suite assembled by Batlle et al. [3] from the problems of de Hoop et al. [8] and Lu et al. [17]. Table 1 collects the published numbers and ours.

The two problems studied in depth sit at opposite ends of the neural-versus-kernel spectrum, and the pair is the point. On structural mechanics the families tie: the pipeline reaches $4.572 \pm 0.010\%$ under the 20000-sample protocol, replicated at ten independently seeded runs, which is below every published method but PARA-Net's 4.55% and within one standard error of that one; and 5.433% under the 1250-sample protocol against a published best of 6.49%. What the coupling adds there is mostly understanding, and the replication is what licenses the reading. The residuals of every architecture we train correlate between 0.83 and 0.97 at every seed; their shared component concentrates where the finite element data are least reliable and is flat in ensemble diversity and training-set size; the hardest cases are hard for all members at once, so averaging does not touch them; and the spread of our own pipeline across seeds is 0.010 points, some sixty times smaller than the 0.65 points that separate the published architectures. The plateau near 4.5% therefore reads as a property of the benchmark's data rather than of any surrogate class, and it is not run-to-run noise that makes it look flat (Section 4.6). Absent regenerated finite-element data this remains an inference, but it is the reading every measurement we made supports. On the rest of the suite the same recipe sorts the problems by the same logic: where the map is smooth and data are plentiful (Helmholtz, Navier–Stokes, Darcy) the kernel stage wins the pipeline's internal selection and lands within a factor of two of the tuned per-problem kernels of Batlle et al. [3], and on Burgers the corrected mean sits near the Fourier neural operator.

On OCO-2 the same components separate by an order of magnitude, and the pipeline's job changes: not to couple two comparable models but to move the kernel's exactness onto the network's representation. Per spectral band of the satellite, the task maps a reduced atmospheric state to a reduced radiance spectrum, and the baseline is the kernel-flow emulator of Lamminpää et al. [13], whose stored predictions we score on the same test points. An exact Matérn head on the trained network's features reaches 4.1% where the same kernel on the raw state reaches 40%; the effective dimension of the two Gram matrices is unchanged, the target's squared native-space norm falls by a factor of about forty, and Proposition 6.16 gives the mechanism. Trained in a loss weighted towards the reported metric and combined per output coordinate, the result improves on the emulator on both of that problem's error metrics at once on all three bands, at ten of ten seeds on two of them and nine of ten on the third (Section 5 is precise about the margins).

Each stage of the pipeline carries one supporting result, and the objects the results cover are the deployed estimators rather than idealizations of them: a symmetrization lemma behind the reflection averaging; a second-moment identity that predicts stacking outcomes from measured residual correlations, sharpened to a two-sided bracket that certifies, from the same measurements, how close any convex ensemble must sit to its floor; finite-sample oracle inequalities for the validated simplex stack, for the per-pixel affine stack the pipeline actually fits, and for the per-coordinate selection that builds the OCO-2 surrogate, chained through the grid-tuned correction into a certificate for the shipped object whose every constant is measured and released; an optimal-recovery bound $\|G - m\|_K \, P_\lambda(u)$ for the corrected surrogate; a pullback identity and a rate for the feature kernel; and a distribution-free coverage statement whose hypotheses the experiment satisfies exactly, the calibration set being disjoint from every selection. The coverage statement matters because the design factor P_λ, though a valid bound, turns out to be a poor pointwise ranking of the actual error, and its split-conformal rescaling is the only uncertainty signal that survives our tests (Section 4.8). Every stated result is also checked numerically: identities to machine precision, inequalities on constructions whose ground truth is known exactly, and the pipeline-level statements on the released artifacts of every seed (Appendix C). The two-member stacking rule in particular is scored prospectively, on every member pair of every seed rather than on selected examples, and that scoring is reported as it came out: the rule is reliable outside a margin band set by its own estimation noise and no better than chance inside it, which is a limitation of the measurement we can state precisely rather than one we can remove (Section 5.2). The symmetrization inequality, which is a statement about an expectation, is reported with the one finite-sample reversal that ten seeds turn up.

Section 2 describes the problems, protocols, and published results. Section 3 specifies the pipeline. Sections 4 and 5 report the two studies, including the data-scaling experiments. Section 6 states the supporting theory, one result per stage, with proofs in Appendix A and their numerical verification in Appendix C, and Section 7 discusses limitations.

2 Benchmark, protocol, and related work

2.1 Problem and data

The dataset originates in the cost-accuracy study of de Hoop et al. [8] and is the one distributed with Batlle et al. [3] and Mora et al. [20]. An isotropic elastic plate occupies $D = (0, 1)^2$; the displacement field w satisfies $\nabla \cdot \sigma = 0$ with a fixed constitutive law, displacement conditions on the bottom and lateral parts of the boundary, and a prescribed normal traction $\bar{t}$ on the top edge $\Gamma_t = [0,1] \times \{1\}$. The quantity of interest is the von Mises stress field σ_v on D. The learning task is the map $G : \bar{t} \mapsto \sigma_v$. Loads are drawn from the Gaussian field $\mathcal{GP}\big(100,\, 400^2(-\Delta + 3^2 I)^{-1}\big)$ with homogeneous Neumann boundary conditions for the Laplacian; outputs are finite element solutions interpolated to a regular 41×41 grid, and the load is sampled at 41 points. The distributed file contains 40000 input/output pairs; the input array stores the load broadcast along the second grid coordinate, which we verified is an exact copy (maximal deviation 0 across all 40000 samples), so all our methods consume the load as a vector in $\mathbb{R}^{41}$.

Errors are mean relative L^2 errors over the test set,

$$\frac{1}{N_{\text{test}}} \sum_n \frac{\|\widehat{G}(u_n) - G(u_n)\|_2}{\|G(u_n)\|_2},$$

computed on grid values in double precision. Quadrature weighting (trapezoidal instead of plain vector norms) changes our reported numbers by less than 0.02 percentage points, and we report the plain-norm convention of the prior work.

2.2 Protocols and published results

Two regimes appear in the literature, and we follow both. In the *high-data* protocol the first 20000 samples are available for training and the last 20000 form the test set; de Hoop et al. [8] report, at 20000 training samples, 4.55% (PARA-Net), 4.67% (PCA-Net), 4.76% (FNO) and 5.20% (DeepONet), and Batlle et al. [3] report 5.18% for their optimal-recovery kernel method (Matérn/rational quadratic; 27.11% for the linear kernel) on the same task. Within the 20000-sample budget we hold out the last 1000 samples (of a fixed permutation) for model selection and stacking, and train on the remaining 19000, so no method of ours sees more than the 20000 training samples used by the baselines. In the *low-data* protocol of Mora et al. [20], 1250 samples are available and the same 20000-sample test set is used; their table gives 8.70% (DeepONet), 6.62% (FNO), 6.95% (the kernel method of Batlle et al. 3), 6.74% (their zero-mean GP), 7.12% (DeepONet-mean GP) and 6.49% (FNO-mean GP). In this regime we carve the validation split (250 samples) out of the 1250, so training uses 1000 samples and every choice made by the pipeline is informed by the 1250 available labels only.

2.3 A data-driven check of the mirror symmetry

The continuous problem is invariant under $x_1 \mapsto 1 - x_1$: the domain and boundary partition are symmetric, and the input law has a symmetric covariance. On the grid, reflecting the load (S) should reflect the stress field along the first grid coordinate (T), i.e. $G(Su) = TG(u)$. Rather than assume this, we test it on the data. For each of the first 200 samples we searched all 40000 loads for the nearest neighbor of the reflected load Su_i and compared the corresponding outputs. For the five best-matching pairs, the input mismatch $\|Su_i - u_j\|/\|u_j\|$ ranges over 0.27–0.32, while the output mismatch $\|Tv_i - v_j\|/\|v_j\|$ ranges over 0.085–0.14; reflecting along the second coordinate instead gives output mismatches of order one. Outputs of near-mirror inputs are far closer than the inputs themselves, which is what equivariance plus a Lipschitz solution map predicts, and the effect singles out the correct output reflection axis. A complementary check on the input law: the empirical mean and covariance of the training loads are invariant under reflection to within 1.1% and 1.5% respectively, as required for Proposition 6.1. Consistently with all of this, reflection averaging at test time improves twenty-nine of the thirty reflected members of the seed campaign (Section 4), which is what Proposition 6.1 predicts: it constrains the expected loss, so a finite test sample is free to go the other way, and at one seed it does.

2.4 Related work

The benchmark sits at the meeting point of three lines of work. *Neural operators* learn maps between function spaces: DeepONet [16] with its branch/trunk factorization, the Fourier neural operator [15] and the broader neural-operator framework [11], and the reduced-basis PCA-Net and PARA-Net architectures assembled for the cost–accuracy study of de Hoop et al. [8]. These methods are fast and mesh-flexible but do not by themselves come with error control, and it is their numbers on this problem that define the plateau we start from. *Kernel and Gaussian process methods* for operator learning approach the same maps through reproducing kernels: the optimal-recovery framework of Batlle et al. [3], with its convergence theory and a-priori bounds, is the most direct comparison, and it is competitive with neural operators on most of the benchmarks it considers. Data-adapted kernels learned by cross-validation [7] and vector-valued kernel formulations extend the reach of this family. *Hybrids* that place a neural network and a kernel in the same estimator are the closest relatives of our pipeline: Mora et al. [20] use a neural operator as the mean of a Gaussian process and fit the two together, and this is the work we most directly build on, differing in that we fit the mean first and the kernel after, tune by cross-validation rather than marginal likelihood, and solve the correction exactly at nineteen thousand points.

Two further threads inform the design. The kernel-flow method [22] learns a kernel from data by minimizing a half-sample cross-validation loss, and its use as a regularizer for the inner layers of a network [33] is exactly the term we analyze in Proposition 6.2; kernel flows have since been used at scale as emulators for physical forward models, for instance in atmospheric radiative transfer retrievals [13] and in the inference of convective-storm structure from satellite observations [24], where presenting the input in its physical parametrization and adapting the kernel to data are what make the emulator accurate. Our pipeline follows the same instinct, with the smoothing of the target performed by a neural ensemble rather than by a hand-chosen reduction. Finally, the effective-dimension reading of the kernel solve (Lemma 6.20) draws on the random-matrix description of kernel Gram spectra [10] and on the classical Marčenko–Pastur [19] and spiked-covariance [2] laws; the same effective dimension governs the generalization of kernel ridge regression [6].

3 Method

The pipeline has three stages: neural means, stacking, and kernel corrections. All components operate on the load vector $u \in \mathbb{R}^{41}$ (standardized by training statistics) and produce stress fields in $\mathbb{R}^{41\times 41}$; all networks are trained with the reported metric as the loss, i.e. the per-sample relative L^2 error in original units, which we found mildly but consistently better than mean squared error on normalized targets.

3.1 Neural means

Residual MLP. The primary mean is a residual multilayer perceptron: three or five residual SiLU blocks of width 1024–1536 mapping $\mathbb{R}^{41} \to \mathbb{R}^{1681}$ directly. This is essentially PARA-Net [8] with a modern training recipe, and it also supplies the penultimate features used by the feature-space kernel stage.

Kernel-conditioned refiner. The second mean is the same residual network given an extra input channel: the kernel method's predicted stress field for the same load, concatenated with the load. It therefore learns to correct the kernel predictor rather than to reproduce the map from scratch, and it is the strongest single model in our study. During training the refiner reads *out-of-fold* kernel predictions (four-fold, so the kernel channel is never fit on the target sample), and at test time the full-data kernel prediction; this keeps the channel honest.

Other instances. The mean slot is not tied to a particular architecture. We also use a Fourier neural operator [15] that consumes the broadcast load as a field, a UNet on the same field representation, and an MSE-trained variant of the MLP; all are drop-in means, all enter the ensemble of Section 4.6, and their pairwise error correlations are one of the measurements of this paper. We additionally implement an encoder–decoder transformer [9, 27] that tokenizes the 41 load samples and decodes the field by cross-attention at the grid points; it is a natural fourth architecture family, but we were unable to train it to the level of the others on the hardware available for this study and report the ensemble without it (Section 4.6 returns to this).

All means are trained with AdamW and a cosine schedule, batches of 128–256, for up to a few hundred epochs (Appendix B lists every hyperparameter). During training each batch is reflected with probability 1/2 (load reversed, target field reflected along the first grid coordinate); at test time each model is evaluated as $\frac{1}{2}(f(u) + Tf(Su))$. Proposition 6.1 guarantees the test-time step cannot hurt on average, and the ablation confirms small consistent gains from both steps. Optionally, the training loss carries a kernel-flow term $\beta\, e_2$ computed from the batch (Section 6.2) with a Gaussian kernel on pooled penultimate features whose log-bandwidth is trained jointly; this variant appears in the ablation.

3.2 Stacking

With M trained models $f_1, \dots, f_M$ we form the convex combination $m(u) = \sum_m w_m f_m(u)$, $w \in \Delta^{M-1}$, minimizing the validation relative error; the weights are initialized at the simplex minimizer of the measured second-moment matrix of Proposition 6.7 and polished by a short search on the 1000 held-out samples (250 in the low-data protocol). A per-pixel variant allows the weights (plus an intercept) to vary over the grid, ridge-fitted on half the validation split and accepted only when it beats the global weights on the other half; it is the variant the final pipeline uses. Stacking on a held-out split rather than uniform averaging costs nothing, guards against a weak member [32], and is statistically almost free at this size (Proposition 6.3).

3.3 Kernel corrections

The stacked mean is then corrected by kernel ridge regression of its residuals. Let $X = (u_1, \dots, u_n)$ be the training loads (standardized coordinatewise), $R \in \mathbb{R}^{n \times 1681}$ the matrix of training residuals $v_i - m(u_i)$, and k a Matérn-5/2 kernel $k(u, u') = \kappa\big(\|u - u'\|/s\big)$. The correction is

$$\widehat{r}(u) = k(u, X)\,\alpha, \qquad \alpha = (K + n\lambda I)^{-1} R,$$

with (s, λ) chosen on the validation split from a small grid (s as a multiple of the median pairwise distance, $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$; the grid is searched on an 8000-sample subsample and the chosen pair is refit on all of X). The corrected surrogate is $m + \widehat{r}$. A second corrective stage repeats the construction with the kernel evaluated on deep features, the concatenated penultimate

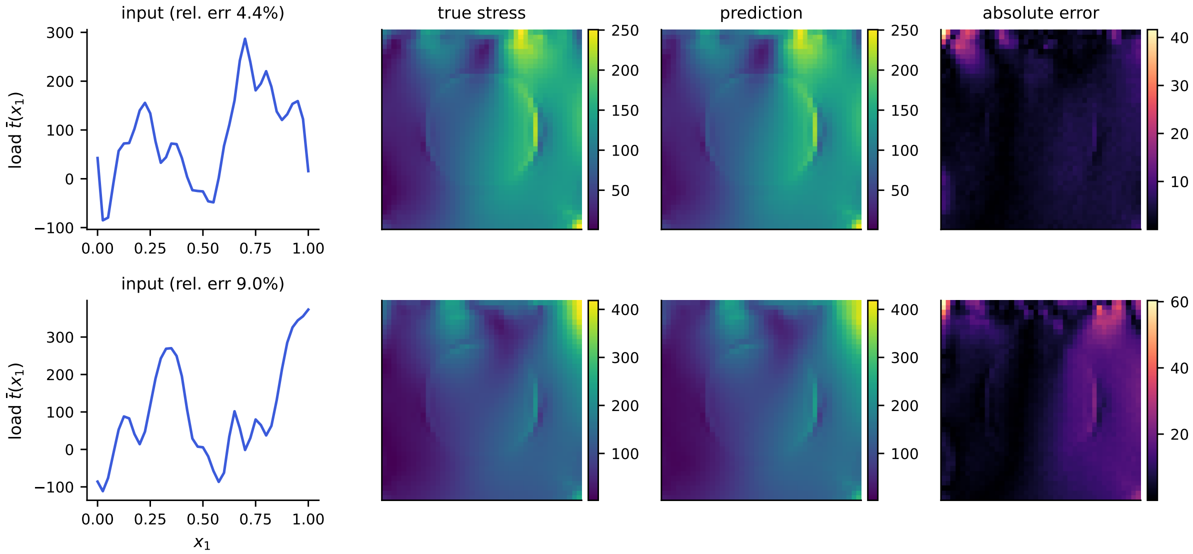

Figure 1: Example predictions of the full pipeline. Top: a median-error test case; bottom: a case at the 98th error percentile. Columns: input load, true von Mises stress, prediction, and pointwise absolute error (note the smaller scale). Errors concentrate in the high-traction corner, as also reported by Mora et al. [20].

activations of the ensemble members (reflection-averaged, standardized), fitted to the residuals of $m + \widehat{r}$; it contributes a smaller improvement and is included when it helps on validation. At $n = 19000$ the exact solve is a single Cholesky factorization of a 19000×19000 matrix, under half a minute in double precision on a laptop CPU (measured in Section 4.10), and prediction is two matrix products; no inducing points, preconditioners, or stochastic solvers are involved. The Gaussian process reading of the same formulas supplies the posterior standard deviation $P_\lambda(u)$ of (2) at negligible extra cost (one triangular solve per evaluation batch), which is the uncertainty estimate studied in Section 4 and certified by Theorem 6.17.

Two remarks on scope. Nothing in the construction uses properties of this particular PDE beyond the verified reflection symmetry, so the recipe (native input parametrization, symmetrized accurate means, validated kernel correction of the residual, posterior standard deviation as certificate) transfers to other operator learning problems with low-dimensional input parametrizations. And the pipeline degrades gracefully: dropping the feature stage, the stacking, or the corrections recovers progressively simpler methods whose individual numbers appear in the ablation table.

4 Results on the structural-mechanics benchmark

4.1 What is replicated, and at how many seeds

Every structural-mechanics number in this section is reported over ten seeds unless it is marked otherwise. A seed here is not a reinitialization of one network: it redraws the 1000-sample validation split from the training pool and reinitializes every trainable member, so the seed spread contains both the model-selection noise and the optimization noise. The test block of 20000 samples is fixed by the benchmark's convention and never moves, at any seed.

Five members are available at all ten seeds. Four of them completed training outright: the kernel ridge on loads, the residual MLP trained on the reported metric, the same architecture trained on normalized mean squared error, and the kernel-conditioned refiner. The fifth, a Fourier neural operator, is finalized from the best-validation checkpoint its trainer wrote before the campaign's wall clock ended the run; nothing is retrained, and Appendix B reports how far each of those runs got and what the truncation cost. The stack and the residual correction are then run at all ten seeds from those five, and so are the second-moment measurement and the conformal calibration.

The sixth member of the configuration this paper originally reported, a UNet, is scheduled after the FNO, and the campaign's wall clock stopped most of its seeds before it began: it completed at two of the ten. That is why the tables below carry five members and not six. The two seeds it did finish are worth reporting rather than discarding, and Table 2 does so. In both, the UNet is the most accurate single member (4.640% and 4.650%, ahead of the normalized-MSE network's 4.742% and 4.731%), and the finished six-member pipeline reaches 4.539% and 4.537% — on both sides of nothing, since the two runs agree to two thousandths of a point, and both sit just below the best published result. Two seeds cannot carry a claim that ten were required to make elsewhere in this section, so we report them as a pair of runs and rest nothing on them; what they do establish is that the six-member configuration is real and was measured, not that its margin is resolved.

One point about how the five-member set was chosen, since it is exactly the kind of decision that invites a post-hoc reading. It is not a configuration searched for after seeing results: it is the pre-specified six-member configuration minus the one member the campaign cannot supply. Whether five members is preferred to four is then decided on validation, per seed, by the same rule that governs every other stage choice here — and it prefers five at ten seeds out of ten (Section 4.3). The four-member pipeline is reported alongside throughout, because the FNO's admission is a measurement and the reader should see what it was measured against.

The low-data pipeline and the OCO-2 combination have the seed coverage stated where they appear, which is ten in both.

4.2 Main comparison

Table 2 reports the high-data protocol. The published numbers are quoted from de Hoop et al. [8] (as tabulated by Batlle et al. 3) and from Batlle et al. [3]; our rows are computed under the split of Section 2, which gives our methods strictly no more training information than the baselines had. The kernel ridge row reproduces the published kernel result almost exactly (5.197% against 5.18%, and it is the most reproducible row in the table: its seed standard deviation is 0.003 points, three thousandths of a point, because the only randomness it carries is the validation draw). We take that agreement as evidence that the protocols are aligned, so the gap opened by the rest of the pipeline is not an artifact of evaluation choices.

The replicated pipeline reaches $4.572 \pm 0.010\%$, the uncertainty being the spread over seeds. That is below every published method except PARA-Net, and it is not separable from PARA-Net: the difference is 0.022 points against an unpaired standard error of 0.020 (Section 4.4), so about one. We claim a tie there and nothing more — not a win, and not the match-by-assertion that a single run would have supported. Against PCA-Net (4.67%) the separation is 0.098 points, five standard errors, and the margins over the FNO (4.76%), DeepONet (5.20%) and the optimal-recovery kernel (5.18%) are larger still.

It is worth saying plainly what changed in the course of replicating this. The configuration

Table 2: Structural mechanics, high-data protocol (20000 training samples, 20000 test). Mean relative L^2 test error. Rows above the rule are published results on the same data, quoted without uncertainties because none were reported. Rows below are ours: mean ± standard deviation over ten seeds, except the last, which is one configuration at the two seeds at which it completed and is listed as both. The four-member row is the same pipeline with the FNO removed, kept because the FNO's admission is a measurement rather than a design choice (Section 4.3). TTA denotes reflection averaging at test time.

Method	Error (%)	Parameters
DeepONet [8]	5.20	–
FNO [8]	4.76	–
PCA-Net [8]	4.67	–
PARA-Net [8]	4.55	–
Optimal-recovery kernel [3]	5.18	–
Kernel ridge on loads (ours)	5.197 ± 0.003	–
Residual MLP + reflection TTA	4.836 ± 0.018	4.9M
+ affine stack over five members	4.595 ± 0.008	22.9M
+ residual kernel correction	$\mathbf{4.572 \pm 0.010}$	22.9M
the same pipeline without the FNO	4.611 ± 0.005	16.5M
Six-member configuration, two seeds	4.539, 4.537	27.3M

reported here reached 4.55% at a single seed in the original study, and that number was presented as matching PARA-Net. The campaign completed that six-member configuration at two seeds, where it averages 4.538%. Rebuilding it across ten seeds — first without the FNO, which had not finalized, and then with it — gives $4.611 \pm 0.005\%$ and $4.572 \pm 0.010\%$ respectively. The four-member figure sits three standard errors *above* PARA-Net; the five-member figure is level with it. So the original claim survives, but only at the configuration that carries the spectral member, and only as a tie rather than as an equality of the two-decimal digits. Section 4.6 argues the resulting cluster is better read as a floor the data impose than as a coincidence of tuning.

4.3 Members and the stack, across seeds

Table 4 gives the five replicated members and the stages built on them. Four features of it are worth naming.

The ordering of the members is stable at the top and not below it, and the seeds are what separate the two cases. Scored on the reflection-averaged validation predictions that the stack consumes, the normalized-MSE network is the most accurate single member at all ten seeds — not nine, ten. Second place is not stable: the refiner takes it at six seeds and the FNO at four, and on test the two are separated by 0.024 points against the FNO's own seed spread of 0.031. So the first rank survives ten independent redraws of the validation split and ten reinitializations and is not a coincidence of tuning; the second is a coin toss that a single-seed table would have reported as a fact. That contrast is the argument for the seeds, in one measurement.

The FNO is the interesting member, and not because it is accurate. At $4.754 \pm 0.031\%$ it is third on average, behind both residual-MLP variants, though it passes the refiner at three seeds of the ten; its seed spread is three to five times theirs — a consequence of checkpoints early-stopped

Table 3: Low-data protocol (1250 training samples, 20000 test). Published rows from Mora et al. [20]. Our pipeline sees only the 1250 labels (250 of them used as the validation split). The pipeline row is mean ± standard deviation over ten seeds. Which stage the validation split selects varies with the seed — the input-space kernel correction at six seeds, its two-stage form at three, the feature-space one at the remaining seed — so the row is the error of the selected pipeline, not of a fixed final stage. The kernel-ridge row is a single run at the multiscale setting.

Method	Error (%)
DeepONet	8.70
GP, optimal recovery [3]	6.95
Zero-mean GP [20]	6.74
FNO	6.62
FNO-mean GP [20]	6.49
Kernel ridge on loads (ours, multiscale, one seed)	6.66
Full pipeline (ours, ten seeds)	$\mathbf{5.433 \pm 0.093}$

anywhere between epoch 10 and 70 (Appendix B). Yet removing it costs the finished pipeline 0.039 ± 0.011 points, and validation prefers keeping it at ten seeds out of ten. Accuracy is not what a member contributes to a stack; the second-moment analysis of Section 4.6 says what is, and the simplex weights there give the FNO the largest single share despite its being third on the metric.

One consequence of how that member was obtained deserves stating, because it runs in our disfavour and a reader should not have to work it out. The FNO is early-stopped by an external wall clock rather than by its own schedule: its cosine learning rate never annealed, and the one full-schedule run of this architecture we have scored 4.705% against the 4.754% these checkpoints average. If that difference is real, every five-member number in this section is a lower bound on what the configuration delivers when its members are trained to completion, and the tie in Table 2 would become a margin. We report the truncated members because they are what ten seeds actually produced, and we do not claim the improvement that finishing them might buy.

The stages are more reproducible than the members they are built from, and the way the seed spread moves along the chain is itself informative. Equal weighting, which fits nothing, takes the members' spread down to 0.005 points on test — ordinary variance reduction. Each subsequent stage fits something, and each gives a little of that back: fitted global weights 0.006, the affine stack 0.008, the correction 0.010. The end of the chain is both the most accurate stage and the least reproducible one. Fitting is not free, and a stack that is tuned harder is not automatically a stack that reproduces better.

Every stage gain is small and every one is resolved. Pairing within each seed on the test set, equal weighting beats the best single member by 0.064 ± 0.007 points, fitted weights beat equal weighting by 0.021 ± 0.003, the affine stack beats fitted global weights by 0.032 ± 0.005, and the kernel correction beats the stack by 0.024 ± 0.005. All four improve at ten seeds out of ten, as does the end-to-end figure of 0.140 ± 0.014 points off the best member the pipeline contains. Each is a few hundredths of a point, which is precisely the resolution a single seed cannot deliver.

Three selection rules were exercised rather than assumed, and all three are decided without touching the test block. Per-pixel affine weights are fitted on half the validation split and accepted only if they beat global convex weights on the other half: accepted at all ten seeds with five

Table 4: Members and stages under the high-data protocol, mean ± standard deviation over ten seeds. Member errors are reflection-averaged where the member has a reflection; the kernel ridge and the refiner train with the symmetrization already applied. "Fitted" weights are solved on the validation split and read out once on test; equal weights involve no fitting and are reported on both splits.

Member or stage	Family / loss	Test error (%)
Kernel ridge on loads	Matérn-5/2	5.197 ± 0.003
Residual MLP	MLP, metric loss	4.836 ± 0.018
Residual MLP	MLP, normalized MSE	4.712 ± 0.006
Kernel-conditioned refiner	MLP on $(u, \mathrm{KRR}(u))$	4.730 ± 0.006
Fourier neural operator	spectral, metric loss	4.754 ± 0.031
Equal-weight ensemble	no fitting	4.648 ± 0.005
Fitted-weight ensemble	solved on validation	4.627 ± 0.006
Affine stack (per-pixel)	selected on a held-out half	4.595 ± 0.008
+ residual kernel correction	–	$\mathbf{4.572 \pm 0.010}$

members, where with four they were declined at one. The correction is accepted on validation at all ten. And the member set itself is chosen the same way — validation prefers five members to four at ten seeds out of ten, which is what admits the FNO. The test-set consequence of that last decision, 0.039 ± 0.011 points, is read out afterwards and plays no part in making it.

4.4 Which uncertainty is the binding one

Seeds are not the only source of uncertainty in Table 2, and on this benchmark they are not the largest. Every seed is scored on the same fixed 20000-sample test block, so the seed standard deviation measures training and selection variability and nothing else. The test block is itself a sample. Its contribution is the standard error of the mean relative error over those 20000 cases, and at a per-sample standard deviation of 0.0169 that is 0.012 points, slightly larger than the pipeline's seed spread of 0.010. Treating the two as independent, the uncertainty on our reported 4.572% is about 0.016 points, and the seeds contribute a little under half of it. A study that replicates seeds and stops has measured the smaller half and is liable to quote it as though it were the whole.

This decides what the table can and cannot support. Comparisons against a published number are unpaired, because we do not hold those methods' per-sample predictions, and so carry the test-block error twice: about 0.020 points in quadrature. The separation from PCA-Net, 0.098 points, is five of those and real. The separation from PARA-Net is 0.022 points, about one, and is therefore not a separation at all — which is the honest form of the claim this paper originally made by asserting equality of two rounded digits. Both figures are conservative, since scoring on identical test points makes the two errors positively correlated in a way we cannot exploit without the baselines' predictions.

Comparisons inside the pipeline are paired, and pairing is worth far more here than seeds are. Evaluating the corrected surrogate and the best single member on the same cases removes the shared difficulty of each case, and the gain of 0.140 points is then resolved at roughly a hundred standard errors at every seed. The same holds for the FNO's admission: 0.039 points, paired, is

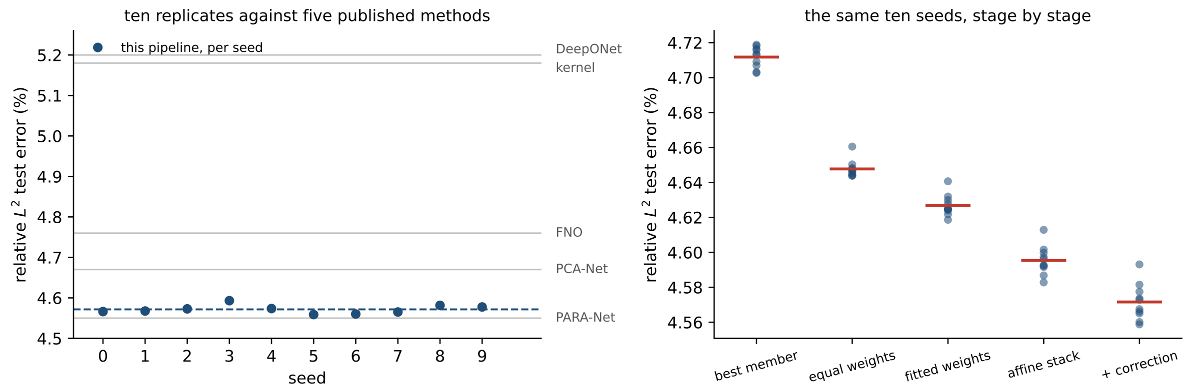

Figure 2: Left: the corrected pipeline's test error at each of the ten seeds (points), their mean (dashed), and the five published single numbers on the same axis (grey). The scatter of one pipeline across seeds is a small fraction of the differences the benchmark is used to adjudicate, which is what licenses reading the published cluster as a property of the data rather than of the architectures. Right: the same ten seeds followed down the pipeline, each stage scored on test, with the stage mean marked. Fitting per-pixel weights lowers the mean and widens the scatter; the correction lowers both.

overwhelming on any one seed, while the interesting quantity — whether it survives retraining — is the across-seed spread of ± 0.011 and the count of ten seeds out of ten. The practical rule the budget implies is the ordinary one, and the campaign is what made it visible: replicate seeds to learn whether a design choice survives retraining, pair on the test set to learn whether it helps, and never use either number to answer the other's question.

4.5 Ablation

Table 5 builds the pipeline up one component at a time. The kernel ridge on loads and the reflection-averaged neural mean start at 5.197% and 4.836%. Stacking is trivial with a single high-data mean and leaves it unchanged; the value of the combination appears once there are several means to combine and again in the correction stage. The feature-space and second input-space corrections did not improve validation error beyond the first correction with a single mean and so were not selected; we report the stages that the validation split actually chose.

Two smaller effects are worth isolating. Reflection test-time averaging lowers the metric-trained MLP's test error by 0.16 points, consistent in direction with Proposition 6.1, which guarantees the sign of the expected change but not its size; the effect is much smaller on the normalized-MSE network (0.02 points) and absent by construction from the refiner, which trains symmetrized. Adding the kernel-flow regularizer (Section 6.2) to the low-data MLP did not help here (it moved test error from 5.64% to 5.87% at a single seed, a comparison the seed campaign did not repeat and which is smaller than the low-data chain's own seed spread of 0.09 points, so we record it as a null result rather than an effect); Proposition 6.2 says what the term estimates, not that estimating it is useful on every problem, and on this smooth-output benchmark the plain metric loss was already a good proxy. Nor does a richer correction kernel help: replacing the single tuned Matérn with a sum of Matérn kernels at several bandwidths leaves the validation error unchanged to two decimals, which is consistent with the residual being

Table 5: Building the high-data pipeline. Mean relative L^2 test error under the 20000-sample protocol; "+" rows are cumulative. The first four rows are ten-seed means; the last two describe the six-member configuration, which completed at two seeds, and give the mean of those two.

Stage	Test error (%)
Kernel ridge on loads	5.197 ± 0.003
Reflection-averaged neural mean	4.836 ± 0.018
+ refiner and MSE variant, affine stack	4.628 ± 0.009
+ residual kernel correction	4.611 ± 0.005
+ FNO member, affine stack	4.595 ± 0.008
+ residual kernel correction	$\mathbf{4.572 \pm 0.010}$
+ UNet member, per-pixel stack (two seeds)	4.557
+ residual kernel correction (two seeds)	4.538

close to kernel-irreducible in the input geometry, and is why the boosted correction stages of Section 3.3 were not selected. The correction's own grid confirms the same reading from the other side: at all ten seeds the tuner selected the largest nugget on its grid, $\lambda = 10^{-3}$, at nine of the ten (the tenth took 10^{-5}), and a length scale of twice the median at five, of the median itself at four and of four times it at one — the most heavily regularized correction available to it, chosen independently at nearly every seed. The length scale is the less stable of the two choices, and the four-member ablation concentrates more tightly than this (eight seeds at twice the median), so the spread here belongs to the five-member configuration.

One positive lesson emerges from the refiner variants. Conditioning the refiner on the kernel method's prediction (test error 4.730%) beats conditioning it on another network's prediction (4.84% at one seed), even though the two conditioning fields have almost the same accuracy; the kernel field carries information complementary to the network's inductive bias, whereas a second network's field is largely redundant. Enlarging the ensemble beyond the residual family is the subject of Section 4.6, where the returns of each addition turn out to be predictable in advance from the correlation structure of the members' errors.

4.6 Architectural diversity and the shared residual

The natural reading of Table 2 is that the remaining error is a modeling problem: different architectures make different mistakes, so a more diverse ensemble should stack to a lower number. We tested this directly, and the test refutes the reading at every seed.

Figure 3 shows the correlation of per-sample normalized residuals between members. Over the ten seeds every pair of members correlates between 0.83 and 0.97, and the kernel predictor — the most alien member by construction, an exact solve on the raw loads rather than a trained network — still correlates 0.86–0.88 with the most accurate network. The mean pairwise correlation is 0.913 ± 0.005 across seeds. The members are not making different mistakes. They are making the same mistake with small private variations, and how small does not depend on the seed.

The second-moment identity of Proposition 6.7 makes the consequence quantitative before any weights are fitted, and the campaign lets us check the prediction prospectively rather than illustrate it once. At each seed we measure the matrix S of normalized residual second moments on the validation split, solve for the optimal simplex weights, and compare the predicted root-mean-square relative error $\sqrt{w^\top S w}$ with what those weights realize. The prediction is $4.940 \pm 0.057\%$ and the realized mean relative error is a factor 0.938 ± 0.003 of it. That factor is the dispersion between a root-mean-square and a mean, and its stability to three thousandths across ten independent replicates is the substantive claim: S alone, measured without ever evaluating the metric, predicts the stack.

The weights it assigns are as informative as the risk it predicts, and they correct something. The optimum spreads across all five members at essentially every seed — refiner 0.31 ± 0.04, FNO 0.34 ± 0.06, normalized-MSE network 0.24 ± 0.08, kernel 0.06 ± 0.03, metric-trained MLP 0.06 ± 0.04 — with the kernel receiving a nonzero share at ten seeds out of ten and the plain MLP at nine. An earlier version of this work reported the kernel and the plain MLP as receiving zero weight. That was an artifact of the solver: the released routine minimized $w^\top S w$ by gradient steps with clipping and renormalization, which is not a projection onto the simplex, and it converged to near-vertex points sitting 0.8–3.6% above the true minimum. Solved properly the answer is a spread, not a vertex, and the qualitative claim that survives is the opposite of a null: every member earns some weight, including the two the broken solver discarded.

That the FNO takes the largest single share while ranking only third on the metric is the whole content of the diversity argument. S is a matrix of second moments, so what it rewards is a residual that is both small and unlike the others', and the spectral member is the only architecture here that is structurally unlike the residual-MLP family. The pairwise admission test of part (i) — a member of error e_2 helps a reference of error e_1 when their correlation falls below e_1/e_2 — admits it at all ten seeds, though narrowly: correlation 0.95–0.97 against a threshold of 0.987–1.001. The kernel is admitted more comfortably, 0.86–0.88 against 0.905–0.918, and also enters the deployed pipeline twice more, through the refiner it conditions and through the affine stack, where weights are not constrained to be nonnegative.

Part (ii) of the proposition puts the infinite-ensemble floor of an equicorrelated family at $\bar{e}\sqrt{\bar{\varrho}}$, which at these measurements is $4.922 \pm 0.056\%$. That number and the pipeline's 4.572% are not on the same footing, and the comparison has to be made carefully for the reason Remark 6.8 gives: the floor is a root-mean-square quantity and 4.572% is a mean, so the mean lies below the floor by Jensen's inequality whatever the estimator does, and a comparison across the two would establish nothing. Converted into the floor's own metric with this pipeline's measured dispersion factor, 4.572% corresponds to about 4.87%, which sits within 0.056 of the floor with the sign undetermined. We therefore claim no margin below the floor here. What can be said without a conversion is the qualitative point, and it is unaffected: the floor bounds convex combinations of the members, while the deployed estimator fits an affine stack per grid point and then corrects its residual with a kernel, neither of which is a convex combination. What the floor measures is how little the convex operation can buy, and the answer, at every seed, is a few hundredths of a point.

Where does the common mistake live? Write r_m for member m's residual field on a validation case and $c = \frac{1}{M}\sum_m r_m$ for the across-member mean, the component that no amount of uniform averaging removes. Measured over the validation split, the shared component carries 90% of the average residual energy (Figure 4). Three of its properties matter. Spatially, its energy concentrates at the two corners of the loaded edge, where the traction boundary condition meets the lateral supports; relative to the local field scale it is three to four times larger there than the grid average, and these are precisely the locations where the finite element solution is least accurate and where interpolation to the 41×41 grid is most strained. Spectrally, it is enriched in high radial frequencies relative to the stress fields themselves: the fields put 97.5% of their energy below radial wavenumber 4, the shared residual only 69%. And its magnitude is statistically independent of everything we can compute from the input: the correlation of $\|c\|$ with the output norm is 0.01, with the load norm 0.01, with the load's total variation 0.01. A component that no architecture avoids, that no input statistic predicts, that lives at the stress concentrations, and that fixed-scale additive noise would reproduce is most economically read as noise of the data-generating process itself, finite element and grid-interpolation error at the singular corners, not as approximation error of the surrogates.

Three independent measurements corroborate this reading. First, the residual kernel correction, which regresses the stack's residual on the load, improves the stack by 0.024 points at $n = 19000$ whatever the kernel scale, and its tuner reaches for the smoothest kernel and the largest nugget on its grid: at this sample size the shared residual is not a function of the load in any way a Matérn RKHS on $\mathbb{R}^{41}$ can see. Second, the error stops responding to data. Under identical recipes at a single seed, the MLP's test error is 4.95% with 3500 training samples, 4.86% with 8500, and 4.86% with 19000 (Figure 6); the ten-seed value at the largest size is $4.836 \pm 0.018\%$, so the last five-and-a-half-fold increase in data buys nothing that seed noise does not already cover. The kernel ridge curve still falls, at roughly $n^{-0.11}$, but toward the same region. Third, and this is what ten seeds add, the error does not respond to reseeding either, and the scale of the comparison is what makes it an argument: the five published architectures in Table 2 span 0.65 points, while ten reseedings of our own pipeline span 0.034, about five percent of it. A modeling limitation should yield to capacity, diversity, data, or luck. This error yields to none of them.

4.7 Where the remaining error is

The seed replicates make it possible to ask not just how large the error is but where it sits, and whether the ensemble touches it. Both questions have the same answer at every seed.

Per sample, the error is concentrated and the concentration is not where an ensemble can help. Ranked by relative error on the validation split, the twenty hardest cases sit at 10.24% against a median of 4.44%, a factor of 2.31. On exactly those cases the five-member ensemble reaches 10.18% and the mean over the individual members is 10.28%: averaging five members, including the architecturally distinct one, recovers a tenth of a point against the average member and six hundredths against the best one, on cases whose error is more than twice the median. Everywhere else the ensemble buys what Section 4.6 says it should; on the hard tail it buys almost nothing, because the members fail there together — and adding the spectral member, which helps everywhere else, does not change that. This distinguishes an irreducible-hardness region of the input space from a shortage of member diversity, and it is the stronger of the two claims to be able to make: adding architectures will not fix these cases.

Spatially, the error is a boundary phenomenon. The root-mean-square pointwise error over the validation samples is 8.99 on the perimeter of the 41×41 grid against 5.64 in the interior, a ratio of 1.59; both figures are stable across seeds to under a percent of themselves. Averaging lowers them to 8.87 and 5.59, so the ensemble takes about a tenth off a boundary error of nine and half that off an interior error of five and a half: it shaves the concentration slightly but does not touch its shape. The location agrees with the anatomy of the shared component above and with the error maps of Figure 1: it is the loaded edge and the supports, where the traction condition meets the geometry, not a diffuse modeling deficit.

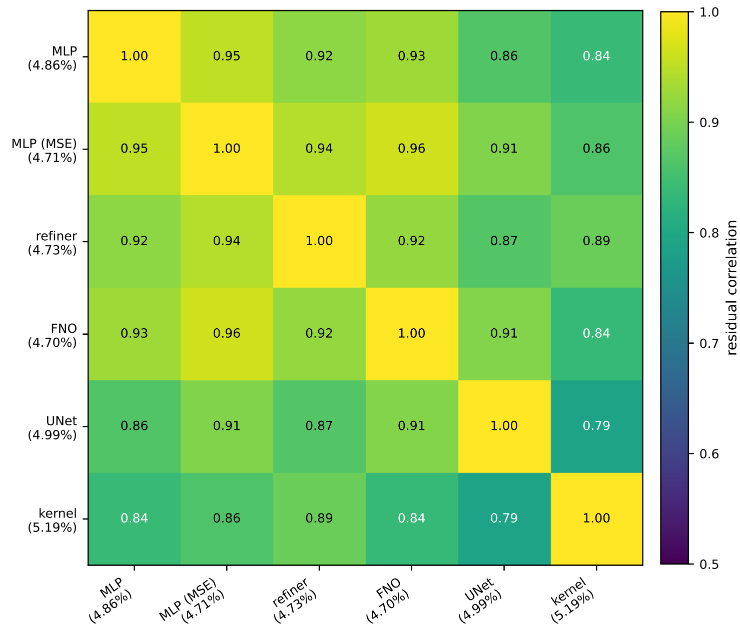

Figure 3: Correlation of per-sample normalized residuals on the validation split, for the six-member configuration at one seed (test-set values agree to two decimals). Across the ten replicated seeds, whose four shared members are a subset of these, every pair correlates between 0.83 and 0.97, with a mean of 0.913 ± 0.005; the kernel predictor, the most alien member by construction, still correlates 0.86–0.88 with the most accurate network at every one of them.

4.8 Uncertainty quantification

Theorem 6.17 certifies the corrected surrogate's error by $\|G - m\|_K\, P_\lambda(u)$, and it is tempting to read the design factor P_λ as a pointwise error bar. On this benchmark that reading fails, in an instructive way. Table 6 confirms the operator factor behaves as the theory predicts: the computable squared RKHS norm of the fitted correction is about $271\times$ smaller when the kernel regresses ensemble residuals than when it regresses raw stress fields, so an accurate mean genuinely leaves a smoother remainder and a tighter certificate (the bound, linear in the norm, by the square root of that). But the design factor P_λ correlates only weakly with the corrected surrogate's absolute error (Pearson 0.08), and *negatively* with the benchmark's relative error. The cause is measurable: P_λ is large for loads far from the training set, those loads tend to have large amplitude, large-amplitude loads produce large-norm stress fields, and dividing by the output norm makes their relative error small. Indeed P_λ correlates 0.65 with the output norm (Figure 5). The error is dominated by the neural mean, whose mistakes are not a function of the input-space geometry that P_λ sees, so a purely input-space power function cannot rank them.

The obvious alternative fares no better, and it fails in the same direction, which is the part worth noticing. Ensemble disagreement, the spread of the members' predictions, is the standard deep-ensemble error signal. Its correlation with the corrected surrogate's absolute error is 0.074 ± 0.006 over the ten seeds, and with the benchmark's relative error it is -0.25 ± 0.04: weakly positive against the quantity the theory speaks about, negative against the quantity the benchmark reports, exactly as for P_λ, and for the same amplitude reason. Two uncertainty signals of quite different provenance — one a Gaussian-process design factor computed from

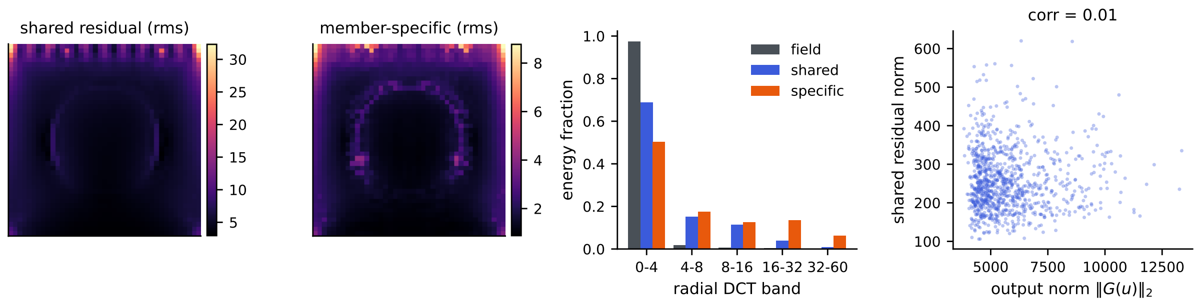

Figure 4: Anatomy of the shared residual c (across-member mean of the residual fields), at one seed. Left to right: rms of c over the grid; rms of the member-specific remainders; radial DCT band energies of the stress fields, of c, and of the specific parts; and the norm of c against the output norm, whose correlation is 0.01.

the inputs alone, the other a spread across trained models — fail on this problem in the same way and to the same degree. Adding architectures does not repair it, and if anything makes it slightly worse: the same correlation is 0.083 on the four-member ensemble and 0.08 on the six-member configuration, so the member whose addition is worth 0.039 points of accuracy buys nothing at all as an uncertainty signal. Section 4.6 says why it cannot: disagreement measures the member-specific components of the error, which carry under a tenth of the residual energy, while the ranking signal for the shared component is invisible to any within-ensemble statistic, because the members agree precisely where they are jointly wrong. Section 4.7 is the same fact seen on the hard tail.

What does hold is coverage, and it holds at every seed. We take the exact split-conformal quantile of the corrected surrogate's error score on a calibration subset drawn from the test block — 1000 calibration cases against 19000 evaluation cases, neither used for training, checkpointing, stacking, or kernel tuning — and evaluate coverage on the disjoint remainder. Conformal validity needs no correlation between the score and the error, only exchangeability, which this protocol supplies exactly (Section 6.7). At the nominal 90% level the observed coverage is $90.2 \pm 0.6\%$, and the right comparison is not with 90% but with the exact finite-sample law: with $m = 1000$ calibration points the coverage of a split-conformal band is $\mathrm{Beta}(901, 100)$ distributed, whose central 95% interval is $[88.1\%, 91.8\%]$. All ten seeds fall inside it. At the 95% level the band is $[93.6\%, 96.3\%]$ and the observed coverage is $95.0 \pm 0.6\%$, again with all ten inside. Scaling the score by ensemble disagreement, which the correlations above say should not help, indeed does not: it changes mean coverage by three tenths of a point and leaves every seed inside the same band. The lesson is narrow but real: a valid pointwise bound (Theorem 6.17) is not automatically a useful pointwise indicator, the standard uncertainty signals can both fail on the same problem, and a relative-error metric has to have its normalization accounted for before a posterior standard deviation means what it appears to.

4.9 Spectra and effective dimension

Figure 7 shows the spectrum of the Matérn Gram matrix on the loads and the effective dimension $d_{\mathrm{eff}}(\lambda)$ computed from it by the exact identity of Lemma 6.20. The spectrum decays quickly: a few hundred eigenvalues carry essentially all the mass, the tail falling many orders of magnitude below the leading modes. At the cross-validated nugget the effective dimension is a few hundred

Table 6: Squared RKHS norm of the fitted kernel interpolant, $\mathrm{tr}(\alpha^\top K\alpha)$, when the same validated kernel and design regress raw stress fields versus ensemble residuals.

Kernel regresses	$\|\widehat{r}\|_K^2$
Raw stress fields ($m = 0$)	1.39×10^{10}
Ensemble residuals	5.14×10^7
Ratio	$271\times$

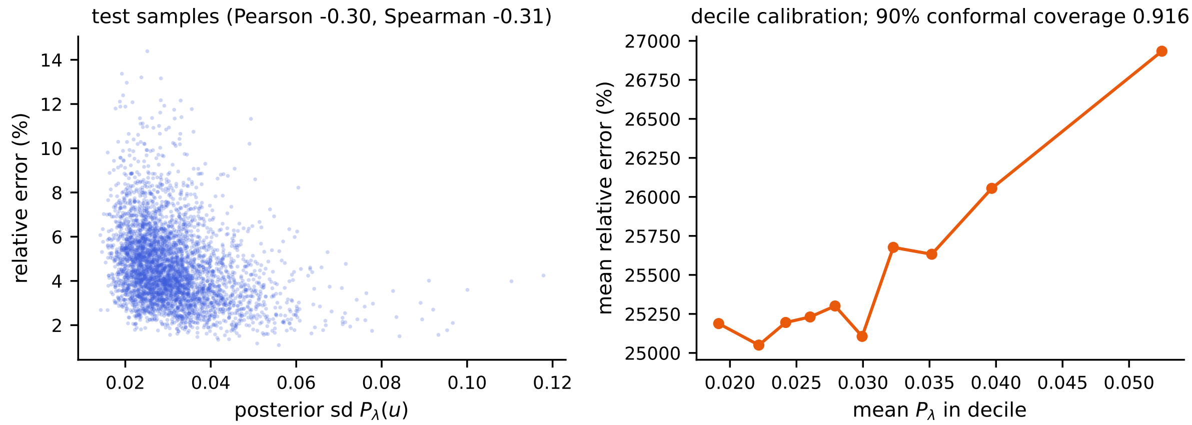

Figure 5: Both panels are one seed. Left: the Gaussian process posterior standard deviation P_λ against the corrected surrogate's relative error on the test set; the association is weak and, because of the output-amplitude confound, slightly negative. Right: decile calibration of P_λ against absolute error, the theory-consistent quantity. Across all ten seeds the distribution-free conformal band attains $90.2 \pm 0.6\%$ coverage at the nominal 90% level, inside the exact finite-sample interval at ten seeds out of ten.

out of $n = 19000$. This is a statistical statement, not a computational one — the dense factorization costs its full n^3 operations regardless — but it is why the correction generalizes despite interpolating nineteen thousand points: the problem the kernel solves has the statistical dimension of the retained spectrum, not of the sample. The cross-validated λ sits at the shoulder of the d_{eff} curve, past the leading modes and into the rapidly decaying tail, matching the reading of Lemma 6.20.

4.10 Cost and reproducibility

The pipeline was built on a single laptop and replicated on four cloud CPU instances. The kernel stages are exact: one Cholesky factorization of the 19000×19000 Matérn Gram matrix in double precision takes 24 seconds on the laptop's CPU (16 threads, OpenBLAS; the matrix itself is 2.9 GB), and between 18 and 32 seconds on the campaign instances, on which it ran ten times, once per seed. Prediction is two matrix products; there are no inducing points, stochastic solvers, or preconditioners, and no GPU is needed for the correction. The factorization performs the full $n^3/3 \approx 2.3 \times 10^{12}$ floating-point operations — numerical low rank does not reduce the work of a dense Cholesky — and the measured rate, about 100 gigaflops per second, is simply what a

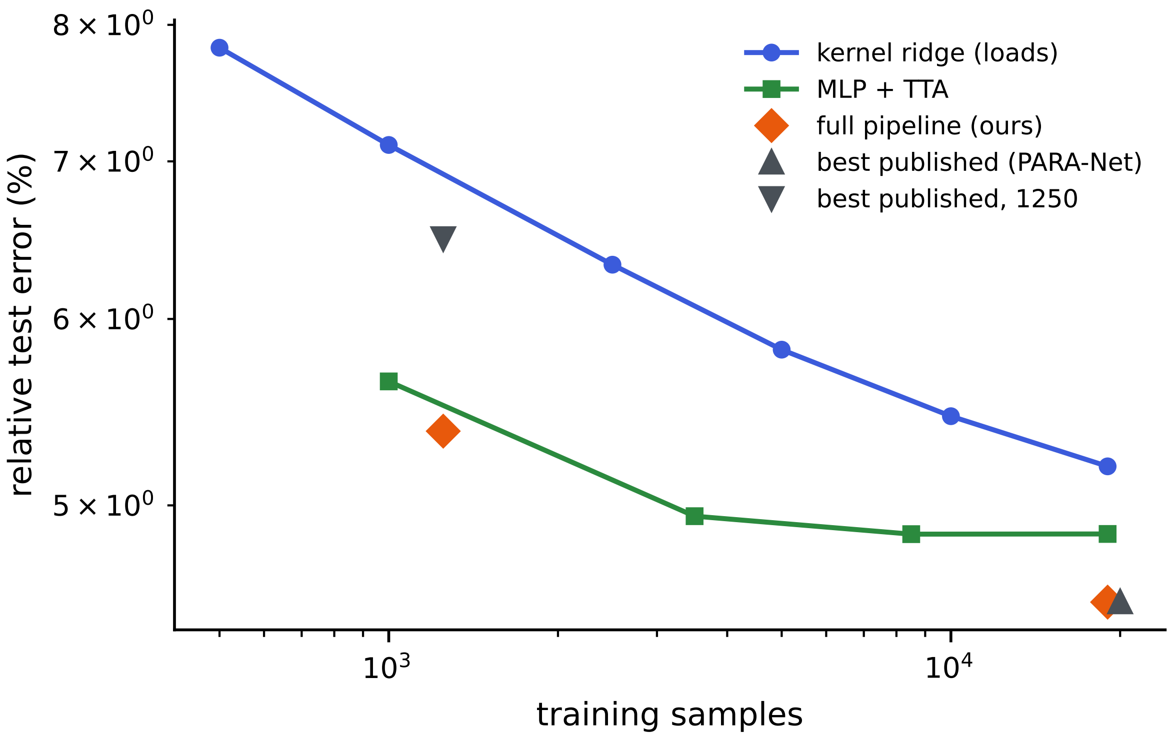

Figure 6: Relative test error against training set size for the kernel ridge on loads (log-log), with the full pipeline and the best published results marked in both regimes.

current multicore CPU delivers; exactness at this n costs half a minute, not a cluster. What the low effective dimension of Section 4.9 buys is statistical rather than computational: it describes where the solve's information concentrates at the working nugget, not its flop count.

Building the Gram matrix, not factoring it, is the stage that needs care at this size: forming it through a full squared-distance matrix and its elementwise transforms allocates several 2.9 GB temporaries at once, which exhausted a 31 GB instance during the campaign. Filling it in row blocks into a single preallocated array holds the peak at the matrix itself and is what the released code does. The neural means are small by the standards of the literature (a few million parameters) and train in the metric itself. Every split is fixed by a published permutation seed, the input is consumed as the 41-dimensional load after the exact broadcast check of Section 2, and every model is selected on the validation split and evaluated once on the test set; we release the splits, the code, and the per-seed result record behind every stage of every run, so any row of any table can be rechecked against the records or regenerated end to end by rerunning the pipeline. The prediction arrays and checkpoints themselves are not distributed, so regenerating a row means retraining it.

5 The OCO-2 radiative-transfer emulator

The structural-mechanics benchmark put the neural mean and the kernel in the balanced regime. To see the other regime we apply the same components to the radiative-transfer emulation problem of Lamminpää et al. [13]: per spectral band of the OCO-2 instrument, learn the map

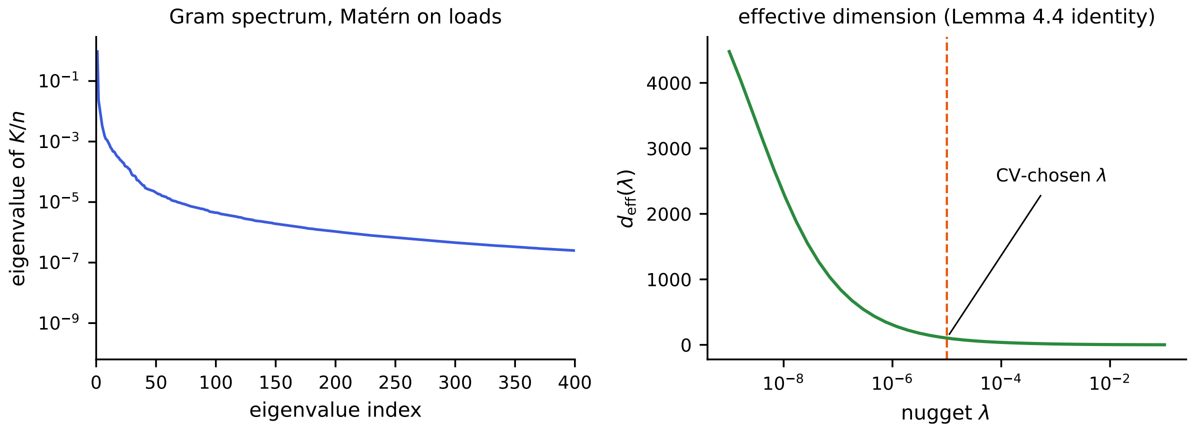

Figure 7: Left: eigenvalues of the Matérn Gram matrix K/n on the loads (log scale); the spectrum decays by many orders of magnitude within a few hundred modes. Right: the effective dimension $d_{\mathrm{eff}}(\lambda)$ from the exact identity of Lemma 6.20, with the cross-validated nugget marked.

from a reduced atmospheric state (20–24 dimensions after the dimension reduction of that paper) to the reduced radiance (40 PCA coefficients of the monochromatic spectrum). The OSF project (`u2t8a`) publishes a training pool of 20000 pairs and, separately, 2000 test states together with the kernel-flow emulator's own predictions on them; we verified the two sets share no point. We split the pool 18000/2000 into training and validation, and every choice in our pipeline — the network checkpoints, the kernel head's length scale and nugget, and the per-coordinate winners of the combination below — is made on that validation split; the 2000 public test states are used once, for the final numbers. The comparison is therefore computed on identical test points against the published emulator itself rather than against our reimplementation of it.

Two error metrics matter, and they disagree in an instructive way. The *reduced* metric is the relative L^2 error on the 40 standardized coefficients. The *radiance* metric maps predictions back to the monochromatic spectrum through the stored PCA projection and norms before comparing; because the projection is orthogonal, the error *numerator* on the reduced side is exactly the diagonally weighted norm $\|s_z \odot (\hat{z} - z)\|$, whose weights concentrate almost entirely on the first few coefficients (the denominator is the full reconstructed-radiance norm, which adds the reconstruction offsets). It is this diagonal weighting of the residual that the per-coordinate combination exploits.

Table 7 carries four findings, and Figure 8 shows the two central ones. First, the representation ladder: the same exact kernel machinery scores 40% on the raw input, 30% after rescaling each input coordinate by its measured sensitivity, and 4.1% on the trained network's features, past the network's own head. The raw-input kernel was limited by its features, not by anything kernel-shaped, and the deep kernel head, an exact solve on learned features, is the strongest single model on the reduced metric. Its margin over the network it feeds on ($4.11 \pm 0.05\%$ against $4.43 \pm 0.05\%$) is small — a third of a point — but it is not a seed accident: the head wins at ten of ten seeds on this band and at ten of ten on each of the other two, paired within seed. We still lean on the ladder's order of magnitude rather than on that gap. Learning the metric rather than setting it by hand does not change this reading. A per-dimension metric fit by the kernel-flows cross-validation loss of Owhadi and Yoo [22] reaches the same 30% on the raw state, and a low-rank Mahalanobis metric with a thousand more parameters does not improve

Table 7: OCO-2 O2 band, scored on the 2000 public test states, which are disjoint from the training pool; all model selection happened on a validation split of the pool. Every trained row is mean $\pm$ standard deviation over ten seeds at a matched 250-epoch budget; the kernel-flow row is the emulator of Lamminpää et al. [13], scored from its own stored predictions on fixed test points, and is identical at every seed. "Features" means the penultimate activations of the trained network; the kernel head is the same exact Matérn solve as everywhere else in this paper, and the two raw-input rows are fit by that same routine under the same budget as the feature heads: the same scale grid $\{0.5, 1, 2, 4\}$ times the median distance, the same nugget grid $\{10^{-8}, 10^{-6}, 10^{-4}\}$, the same 6000-sample tuning subsample, and the winner refit on all 18000 training points. An earlier version of this work fit those rows with a lighter diagnostic script and said so; the rows here are the matched ones, which is why the selected hyperparameters vary from seed to seed. The ARD radiance cell is left empty because validation, which scores the reduced metric, ties between two hyperparameter configurations whose radiance differs by a factor of 2.2 (0.150% at seven seeds, 0.329% at three); a mean over that mixture would describe the selection noise, not the model. Table generated from the campaign artifacts by `campaign/gen_oco_table.py`.

model	reduced	radiance
Matérn kernel, raw input, one length scale	$40.57 \pm 0.10\%$	$0.233 \pm 0.003\%$
Matérn kernel, raw input, sensitivity-scaled (ARD)	$29.98 \pm 0.15\%$	—
kernel-flow emulator [13]	16.89%	0.0448%
residual MLP, flat metric	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
residual MLP, weighted-coefficient loss	$49.32 \pm 0.40\%$	$0.0442 \pm 0.0044\%$
residual MLP, exact radiance loss	$59.02 \pm 0.76\%$	$0.0571 \pm 0.0057\%$
Matérn head on the flat network's features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
Matérn head on the weighted network's features	$21.70 \pm 0.52\%$	$0.0319 \pm 0.0020\%$
Matérn head on the radiance network's features	$20.48 \pm 0.61\%$	$0.0379 \pm 0.0037\%$
per-coordinate combination	$\mathbf{4.12 \pm 0.05\%}$	$\mathbf{0.0294 \pm 0.0020\%}$

on it; on the features, where the network has already conditioned the representation, a learned metric gives no gain over the isotropic kernel. The metric is a constant-factor lever, and the representation is the order-of-magnitude one.

5.1 What the metric buys, measured

Proposition 6.15 splits a metric's advantage into a constant σ_1^s and, only under an approximate-rank gap, a rate. The seed campaign scores that split directly: both raw-input kernel rows are refit at five nested training sizes over a sixteenfold range, at ten seeds, on each of the three bands, with the sensitivity map estimated causally from the first n rows only so that no row sees data a smaller sample would not have had.

Two things come out, and one of them corrects a number reported in an earlier version of this work. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, counting the singular values that hold 99% of the squared energy. The participation ratio is a softer count and agrees on O2 (11.99 against 12) while sitting a third of a unit below the threshold count on the two CO_2 bands (13.63 against 14, 13.33 against 13); the rank ratio is between 0.54 and 0.60 on either reading, which is what the argument uses. That is a rank ratio between 0.54 and 0.60, not the near-full rank previously stated for O2, and it is the regime in

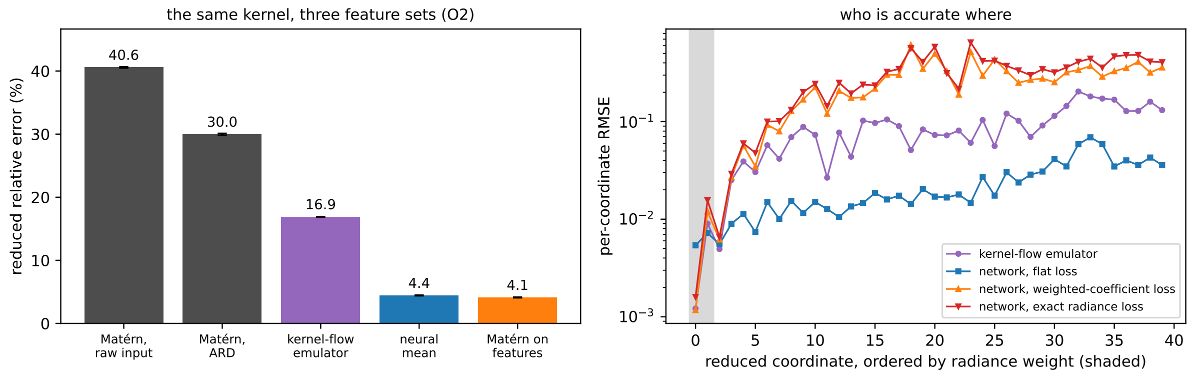

Figure 8: Both panels are drawn from the ten-seed campaign by `campaign/make_oco2_fig.py`: bars and error bars in the left panel are the mean and standard deviation over the ten seeds, curves in the right panel are per-coordinate test RMSE averaged over the same ten. Left: the representation ladder on the O2 band. The same exact Matérn machinery moves from 40% on the raw input to 4.1% on the trained network's features; the kernel-flow emulator and the network sit in between. Right: per-coordinate test RMSE with the reduced coordinates ordered by their radiance weight (shaded). Networks are named by the loss they were trained on. The per-output-tuned kernel-flow emulator is most accurate exactly on the heavily weighted leading coefficient (0.0012 against the flat network's 0.0054), while the flat network is nearly four times more accurate across the trailing thirty coordinates that the radiance barely sees (0.0276 against 0.1037). Both weighted losses buy the leading coefficient at the price of the tail, and by almost the same amount (0.0012 and 0.0016), which is the measurement behind the reading below; the per-coordinate combination takes each coordinate from whichever model wins it.

which the proposition would permit a rate gain and not merely a constant.

The rate gain does not appear. Table 8 gives the paired difference of the two fitted exponents at each band. On O2 the two exponents are indistinguishable across seeds; on the CO_2 bands the adapted exponent is steeper, but by 0.011 and 0.052 against isotropic exponents of 0.086 and 0.203, where the rank ratios above would license something several times larger. What is stable instead is the constant: at the largest training size the isotropic error exceeds the adapted error by a factor of 1.353 ± 0.008, 1.118 ± 0.002 and 1.514 ± 0.114 on the three bands, and those ratios are flat in n over the whole sweep. The reading we take is the one the proposition is stated to support and no more: a metric is worth a constant here, the constant is between 1.1 and 1.5, and the tenfold gap to the feature kernel is not something a metric closes. Both fitted exponents are also steeper than the bound's own s/d, so the absolute rates are not being tested here — only the difference between them, which is what the split concerns.

At its largest size the sweep recovers the raw-kernel rows of Table 7 to the digit, seed by seed and including the selected hyperparameters. We record that as a consistency check on the two code paths and nothing more: both call the same tuning routine on the same standardized inputs, so the agreement is a re-execution of one deterministic computation rather than corroboration by an independent measurement. One protocol note: the table's matched 250-epoch budget is deliberately short of convergence for the network rows. An earlier version of this work trained longer and reached 3.99% (flat network) and 3.82% (feature head) on this band, and a single 750-epoch run of the seeded code reproduces those levels (3.82% and 3.71%, with the combination

Table 8: The raw-input kernel refit at nested training sizes, ten seeds per band. “Rank” counts singular values of the causally estimated sensitivity map holding 99% of the squared energy, out of the input dimension. The exponent difference is paired within seed; its standard deviation is over seeds. The last column is the isotropic-to-adapted error ratio at $n = 18000$.

Band	Rank	−slope iso	−slope adapted	difference	ratio at n=18000
O2	12/20	0.221 ± 0.007	0.215 ± 0.017	$+0.006 \pm 0.015$	1.353 ± 0.008
weak CO_2	14/24	0.086 ± 0.005	0.097 ± 0.006	-0.011 ± 0.009	1.118 ± 0.002
strong CO_2	13/24	0.203 ± 0.029	0.255 ± 0.020	-0.052 ± 0.021	1.514 ± 0.114

at 3.71% reduced and 0.0174% radiance). The budget moves the levels of the trained rows together; at the diagnostic seed it reverses none of the orderings discussed below.

Second, the training metric behaves as a budget, and the campaign now measures which part of the metric carries the budget. The second network in Table 7 is trained on a diagonally weighted loss over the reduced coefficients, $\|s_z \odot (\hat{z} - z)\|/\|s_z \odot z\|$. That is a surrogate for the radiance-relative error, not the error itself: the reported radiance metric maps predictions through the stored PCA reconstruction and divides by the reconstructed-radiance norm, and the surrogate omits the reconstruction. An earlier version of this work described that row as radiance-trained, which overstated it, and drew from the comparison the rule “train in the metric you report”. The campaign adds the row that tests the rule — the third network is trained in the exact radiance metric, reconstruction and denominator included — and the rule does not survive. Trained in its own reported metric, the exact-loss network scores *worse* there than the surrogate-trained one: 0.0571 ± 0.0057% against 0.0442 ± 0.0044% on O2 and 0.093 ± 0.008% against 0.075 ± 0.003% on SCO2, the surrogate ahead at ten of ten seeds on both bands, with WCO2 the reverse at eight of ten (0.064 ± 0.015% against 0.072 ± 0.007%). The kernel heads built on their features repeat the pattern. Tripling the training budget narrows the O2 gap (0.0212% against 0.0201% at 750 epochs, one seed) without reversing it, so slow optimization of the exact loss is at most part of the story. The reason the surrogate gives nothing away is the orthogonality noted above: the two losses share their numerator exactly, and differ only in the per-sample denominator, which reweights states, not coordinates. What the metric buys is coordinate weighting, and either loss buys it — the weighted networks give up the trailing coordinates the radiance barely sees and win the reported metric by a factor of three to four over the flat network, whose per-coordinate profile shows the gap directly: the kernel-flow emulator (lengthscales tuned per output) predicts the leading coefficient to 0.0012 root mean square against the flat network’s 0.0054, while the flat network is three to four times more accurate on the trailing thirty coordinates. The practical rule that survives the measurement is narrower than the one we first drew: train in a loss that concentrates where the reported metric concentrates. Matching the metric’s weighting is what pays; matching its algebra is not.

Third, the same-class floor of the structural-mechanics study reappears here, one level down, and the campaign lets us test it rather than illustrate it. Proposition 6.7 is stated for the squared-metric risk, so the measurement is made there: for each band we form the matrix S of the ten seeded copies of the flat network on the test set, in the relative metric, and read the floor off it. Three things follow, and the third corrects this paper’s earlier reading.

The identity itself holds numerically. At uniform weights $w^\top S w$ and the directly measured risk of the averaged prediction agree to 7×10^{-18}, 3×10^{-17} and 1×10^{-17} on the three bands —

the proposition's content, checked against real predictions rather than assumed.

The floor is where the proposition puts it. Over all 45 seed pairs per band the residual correlations are 0.787 ± 0.010 (O2), 0.971 ± 0.001 (WCO2) and 0.961 ± 0.001 (SCO2), tight enough that the equicorrelated case of part (ii) is a fair description. It predicts a ten-member value of 4.419%, 17.997% and 9.267% against realized 4.419%, 17.997% and 9.267%, and infinite-seed floors of 4.360%, 17.970% and 9.248%. Ten seeds have already taken essentially all of what seed-ensembling has to give: the remaining distance to the infinite-seed floor is six hundredths of a point on O2 and less on the other two bands.

What the floor does not do is fall to a change of member. An earlier version of this work reported that the Matérn head on learned features passed below it. That comparison was invalid: the floor is a squared-metric quantity and the head's error was quoted from Table 7, which reports the mean relative error $\mathbb{E}\,\ell$. Measured in the same metric as the floor, the head reaches 4.575%, 18.11% and 9.326% — better than an individual seeded network (4.915%, 18.238%, 9.433%) and worse than that network's ten-seed ensemble. The ranking is the same in the mean metric, where the seed ensemble reaches 3.916% against the head's 4.111% on O2. At this budget a different kind of member buys less than ensembling ten copies of the plain one, and we say so; the order-of-magnitude result of this section is the representation ladder, which does not depend on this comparison. Ensembling within an architecture class saturates by the same second-moment law on both problems, and the two ways of spending a fixed compute budget — more seeds of one member, or one member of a better kind — are closer here than we previously claimed.

The floor is a property of the architecture class, not of the problem, and it can be probed directly. Different architectures decorrelate: on WCO2 two residual MLPs of different activation correlate at 0.84, a wide shallow network against the residual MLP at 0.40, and a random-Fourier-feature network against it at 0.08 to 0.17, all far below the 0.971 ± 0.001 that ten seeds of one architecture reach on this band. Yet the floor did not move, because its other input is accuracy: by part (i) of Proposition 6.7 a member of error e_2 helps a reference of error e_1 only when their correlation falls below e_1/e_2, and across a bandwidth sweep the Fourier network was either accurate and correlated (error 31%, correlation 0.54, just above the 0.53 threshold) or decorrelated and inaccurate (error above 140%). No architecture we tried was both accurate and decorrelated enough to lower the floor, which is the honest statement of why the residual MLP is the member of record: not that diversity is impossible, but that on this map no more accurate diverse member was found. Figure 9 shows both halves of this: the correlations that make the floor class-specific, and the accuracy-decorrelation trade-off that keeps it in place. Adding the Fourier members to the per-coordinate combination changes the reported error by less than a hundredth of a point, in either direction, on all three bands: the honesty-split blend gives them no weight.

Fourth, the regimes of Section 6.4 are decided by these numbers before any stack is fit. Against the flat network the raw-input kernel has $\varrho = 0.14$ and $e_n/e_k = 0.109$ (the network's 4.43% over the kernel's 40.57%): the condition of Proposition 6.7 fails, by a margin that ten seeds do not put in doubt, and the kernel is dropped — in contrast to structural mechanics where the same test keeps it. Because the radiance metric is diagonal in the reduced coordinates, the per-coordinate combination of Proposition 6.12 is optimal for both metrics simultaneously. It yields a single surrogate that beats the kernel-flow emulator on both metrics at once, now on all three bands: O2 reaches $4.12 \pm 0.05\%$ against 16.89% reduced and $0.0294 \pm 0.0020\%$ against 0.0448% radiance, SCO2 $8.08 \pm 0.01\%$ against 16.14% and $0.0584 \pm 0.0041\%$ against 0.1147%, and WCO2 $16.28 \pm 0.06\%$ against 24.06% and $0.0507 \pm 0.0052\%$ against 0.0599%. Counted by

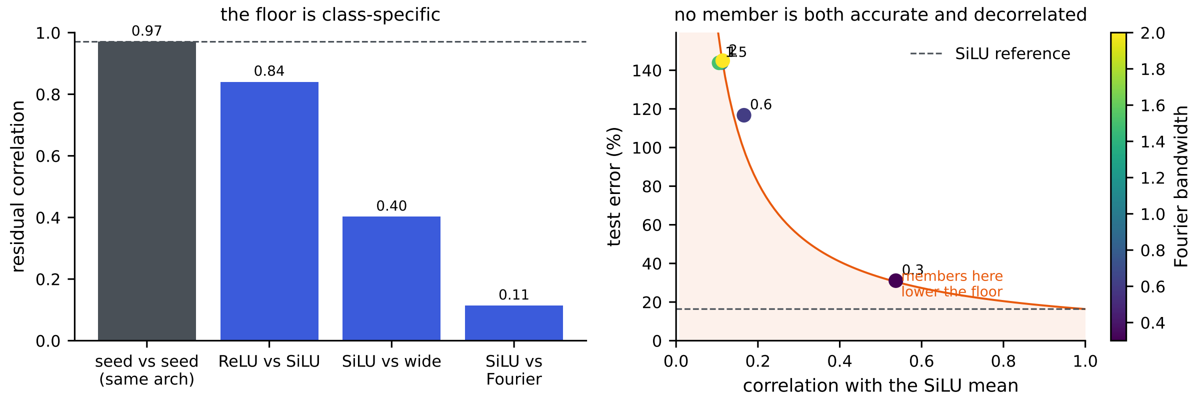

Figure 9: The ensembling floor on WCO2; the architecture comparison in the left panel is drawn from one seed of each, and the seed-pair correlation it shows is the campaign's mean over 45 pairs. Left: residual correlations. Two seeds of one architecture correlate near the value that sets the floor (dashed); different architectures correlate far less, so the floor is a property of the architecture class. Right: the Fourier-feature member across bandwidths. A member helps the ensemble only in the shaded region, below the e_{ref}/ϱ curve of Proposition 6.7(i); the accurate settings are too correlated and the decorrelated settings too inaccurate, so none lands there with room to spare.

seed, the combination beats the emulator on both metrics at ten of ten seeds on O2 and SCO2 and at nine of ten on WCO2, the single exception missing on the radiance metric by 0.0015 points.

WCO2 was a stated exception in an earlier version of this work, where the two best heads split the metrics and no single model won both. What removed it was not a better fit but a wider pool: the combination selects per coordinate from six members here, the flat, weighted and exact-radiance networks and the Matérn head on each of their features, where previously the two radiance-loss members were absent. The emulator's own predictions are not among the members, so the comparison remains against an outside baseline. The band that needed the extra members is the one whose metrics disagreed, which is what the per-coordinate argument predicts; but a nine-of-ten margin of 0.0015 points is thin, and we report it as a band that has just crossed rather than as a comfortable win. The code releases the per-band numbers alongside this paper.

5.2 Where the two-member rule can be trusted

Part (i) of Proposition 6.7 is the one result in this paper used as a decision rule rather than as a description: it says a weaker member helps exactly when $\varrho < e_1/e_2$, and we drop members on that test. A rule of that kind deserves to be scored the way it is used, so we decide it on the validation split and check it on the test block, for every pair of the eight members at every band-seed — $\binom{8}{2} = 28$ pairs at each of the thirty seeds, 840 in all.

Two separate things are being tested by this, and separating them is the point. The first is whether the proposition is true. Evaluated with both sides computed on the same block, the criterion $\varrho < e_1/e_2$ agrees with whether the optimal convex mix actually beats the better member on all 840 pairs, without exception. That is the proposition's own content rather than evidence about the world — it is an identity, and this is the numerical check that we implemented it correctly — but 840 instances is a strong form of that check, and it locates the difficulty elsewhere.

The second is whether the rule survives being used, which requires ϱ and e_1/e_2 to be estimated on data the decision does not see. Decided on validation and checked on test, the rule is correct on 62.7% of the 840 pairs. The average is the wrong summary: sorted by the margin $|\varrho - e_1/e_2|$ that the rule must resolve, the outcome is 105 of 255 below 0.02, 94 of 217 from 0.02 to 0.05, 79 of 86 from 0.05 to 0.10, 217 of 250 from 0.10 to 0.25, and 32 of 32 above 0.25 — that is 41%, 43%, 92%, 87% and 100%. These counts are not independent observations: the 28 pairs of a band-seed share members, and the ten seeds of a band share one test block, so the bins should be read as describing a trend across margin and not as binomial samples. Nothing about the proposition degrades in the small-margin pairs; what degrades is the transfer of two estimates from one block to another.

That transfer is measurable, and it is what sets the band. The signed gap $D = \varrho - e_1/e_2$ moves between validation and test with standard deviation 0.099 across the 840 pairs, essentially unbiased (mean -0.0005). This is an order of magnitude larger than the seed-to-seed scatter of D within a fixed pair, which is 0.010: replication across seeds is not what limits the rule, and a study that varied only the seed would have missed this. With that σ, Proposition 6.9 bounds the disagreement probability by $\exp(-\Delta^2/2\sigma^2)$, which guarantees at least 76% correct at the median margin of the $[0.10, 0.25)$ bin and essentially no errors above 0.25. The measured rates, 87% and 32 of 32, clear that guarantee at every bin and meet it at the top one. Because the pairs are binned by the margin the decision actually sees, it is part (ii) of the proposition that applies, and it carries the same constant.

The practical reading: the rule is dependable when the two quantities are separated by more than about 0.25, suggestive between 0.05 and 0.25, and uninformative below that. This applies to our own use of it. The kernel-versus-network decision earlier in this section has $\varrho = 0.14$ against $e_n/e_k = 0.109$, a margin of 0.031, which falls in the band the scoreboard says not to trust, and we therefore do not rest the exclusion of the raw kernel on that comparison. What does settle it is the same proposition read for magnitude rather than for sign: with these two errors and this correlation, part (i) puts the unconstrained two-member optimum at 4.4299% against the network's own 4.4320%, and the optimizing weight falls just outside the simplex, so no convex mix reaches even that value. Whichever side of the threshold the pair truly falls, the entire quantity in dispute is two thousandths of a point, so the member is dropped on the size of the available gain and not on a threshold the data cannot resolve. Reporting the margin alongside the verdict is what makes that distinction visible, and we recommend it wherever this rule is used.

5.3 The error is limited by data, and yields to more of it

The structural-mechanics error was flat in the sample size because it is set by the noise of the finite element data; the emulator error is not, and the contrast is the sharpest practical difference between the two problems. The left panel of Figure 10 fixes the O2 task and varies only the number of training pairs, scoring each on a disjoint held-out set. The corrected surrogate's reduced-radiance error falls as a clean power law in n, fitted slope -0.68, from 74% at a few hundred pairs to 4.3% at $n = 17700$, still descending at the largest sample size we can afford and approaching the 3.8% our full pipeline reaches on all 18000 pairs (Table 7). The exact kernel on the raw state improves far more slowly, so at these sizes the network is the component that turns data into accuracy. The residual on this task is sample-limited rather than noise-limited:

unlike the structural-mechanics floor it recedes as pairs are added, so the largest lever on this problem is simply more of them, and pairs are cheap because the forward model is a program. The complementary case is the full-resolution $58 \to 3048$ map, where the error is instead flat in n across the few hundred retrievals we hold: that task is limited by its representation, not its sample count, and the reduction the emulator applies is what moves it.

A second benchmark carries the same reading into a data range OCO-2 cannot reach, though only over part of the range we first attempted, and the reason is worth stating before the result. ClimSim [34] is a climate emulation dataset of about ten million samples that maps a 124-dimensional atmospheric state to 128 physics tendencies, with published neural baselines. An earlier version of this work swept the training size across more than three orders of magnitude here and reported a crossover between the two model families. That sweep selected the kernel's hyperparameters against test rows. We reran it under the protocol used everywhere else in this paper — selection on a held-out block drawn from the training pool, the test block scored once — and the rerun reached only the three largest sizes before the campaign ended. The small-sample points are therefore withdrawn rather than restated, the superseded script is kept in the release so the difference can be examined, and the right panel of Figure 10 plots the leak-free points alone. ClimSim publishes no canonical test split, so each seed there draws its own 20000-row test block and the spreads below include that variation.

What the leak-free points establish is the high-data half of the picture, and they establish it cleanly. The exact kernel on the state is flat: 0.390 ± 0.007, 0.391 ± 0.006 and 0.387 ± 0.007 at one, two and three and a half million samples, unmoved across a factor of three and a half in data. The neural mean over the same range is both far above it and still climbing: 0.544 ± 0.005 at one million (five seeds), 0.563 ± 0.004 at two million and 0.575 ± 0.004 at three and a half million (three seeds each), closing on the published multilayer-perceptron baseline near 0.6. The separation is some twenty times the seed scatter at every size, so the ordering at these sizes is not a matter of which run one looks at, and the contrast between a data-limited mean and a saturated kernel is exactly the one the OCO-2 sweep shows on its own scale.

What we do not now claim is where the two families cross. That was the striking part of the earlier figure and it rested entirely on the withdrawn small-sample points; the leak-free sweep begins above any plausible crossing and so cannot locate it. The regimes of Section 6.4 still gain a data axis from what remains — an accurate mean is worth building once there is enough data to train it past the kernel, and at these sizes there plainly is — but the sample size at which that becomes true is an open number here, and rerunning the small-n ladder under the corrected protocol is the obvious next measurement.

6 Theory

Throughout this section $\mathcal{U} = \mathbb{R}^p$ denotes the (discretized) input space and $\mathcal{V} = \mathbb{R}^q$ the output space, $G : \mathcal{U} \to \mathcal{V}$ the target operator, μ the input distribution, and $(u_1, v_1), \ldots, (u_N, v_N)$ i.i.d. draws with $v_n = G(u_n)$; the data carry no sampling noise in the usual sense, coming from a deterministic solver, though Section 4.6 has something to say about solver error. We write $\ell(w, v) = \|w - v\|_2 / \|v\|_2$ for the relative error and $R(\widehat{G}) = \mathbb{E}_{u \sim \mu}\, \ell(\widehat{G}(u), G(u))$ for the risk, which is the quantity reported in Section 4. The results below are largely elementary, but each one licenses a specific design decision in the pipeline of Section 3, and each has a measurable footprint in the experiments: one statement per stage, from the symmetry handling and the kernel-flow term through stacking, the residual correction, its coverage guarantee, and the choice of nugget.

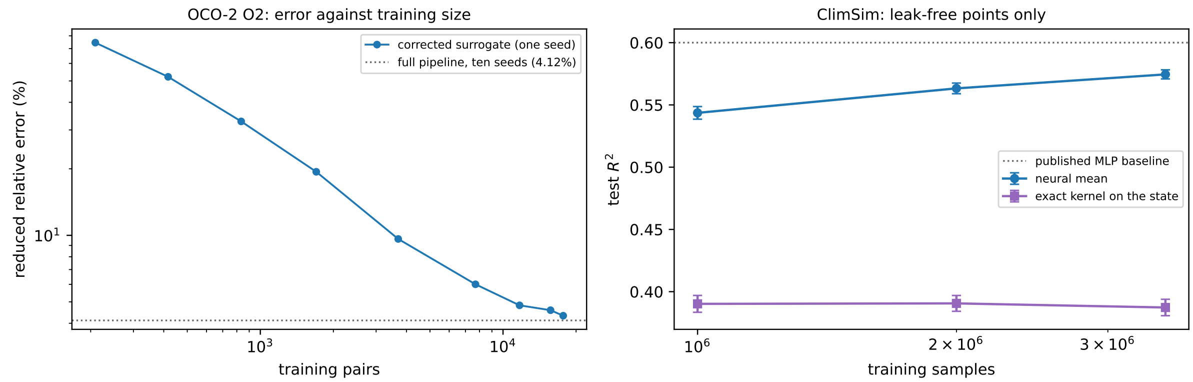

Figure 10: Error against training-set size, drawn by `campaign/make_scaling_fig.py`. Left: the OCO-2 O2 task, reduced error of the corrected surrogate at one seed, from the data-scaling run; it falls as a power law (fitted slope -0.68) to 4.3% at $n = 17700$, against the 4.12% the ten-seed pipeline reaches on all 18000 pairs (dotted). Right: ClimSim, test R^2 over the outputs with non-negligible variance, showing only the points measured under the corrected selection protocol — five seeds at one million samples and three at each of two and three and a half million, error bars over seeds. The small-sample points of the earlier version of this figure are withdrawn, so the crossing between the two families is not shown and is not claimed; over the range that remains the kernel is flat and the neural mean is far above it and still rising toward the published baseline (dotted).

Proofs are collected in Appendix A.

6.1 Symmetry averaging does not increase the risk

The elasticity problem behind the benchmark is invariant under the reflection $x_1 \mapsto 1 - x_1$: the domain, the boundary partition, and the constitutive model are all symmetric, the input distribution has a reflection-invariant covariance, and reflecting the load reflects the stress field. On the discrete grid this is expressed by two permutation matrices: $S \in \mathbb{R}^{p\times p}$ reverses the load samples and $T \in \mathbb{R}^{q\times q}$ reverses the rows of the stress field. Both are orthogonal involutions. Section 2 describes a direct data-driven check of the equivariance $G(Su) = T\,G(u)$; we treat it as an assumption here.

Proposition 6.1 (symmetrization)**.** *Assume $G(Su) = TG(u)$ for μ-a.e. u, that μ is invariant under S, and that T is an orthogonal involution ($T^2 = I$). For a measurable $\widehat{G} : \mathcal{U} \to \mathcal{V}$ define its symmetrization*

$$\widehat{G}_S(u) \;=\; \tfrac{1}{2}\left(\widehat{G}(u) + T\,\widehat{G}(Su)\right).$$

Then, for every loss $\ell(\cdot, v)$ that is convex in its first argument and satisfies $\ell(Tw, Tv) = \ell(w, v)$,

$$\mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}(u), G(u)\big).$$

The relative error satisfies both hypotheses (T orthogonal gives $\ell(Tw, Tv) = \ell(w, v)$). Proposition 6.1 justifies the two uses of the symmetry in the pipeline: averaging each trained model with its reflected evaluation at test time can only help on average, and training on reflection-augmented data is ordinary empirical risk minimization for the symmetrized class. The measured effect is small and, being an expectation, need not hold sample by sample: over the thirty reflected members of the seed campaign it helps twenty-nine times and hurts once, by a thousandth of a point (Appendix C). The convexity argument behind the proposition is the standard one for group-averaged predictors; Lyle et al. [18] give a general account.

6.2 What the kernel-flow regularizer estimates

The kernel-flow (KF) loss of Owhadi and Yoo [22], in the ℓ^2 variant used by Yoo and Owhadi [33] to regularize deep networks, enters our ablation study as an optional term in the training of the neural means. Its motivation is often stated informally ("a kernel is good if half the data predicts the rest"). The following observation makes the statement exact. Let $z_i = \phi_\theta(u_i)$ be the features of a batch $B = \{1, \dots, b\}$ (b even), let k be a positive definite kernel on feature space, and for $C \subset B$, $|C| = b/2$, let I_C denote the k-interpolant of $\{(z_j, v_j) : j \in C\}$. The batch KF loss is

$$e_2(\theta; C) \;=\; \sum_{i \in B} \bigl\| v_i - I_C(z_i) \bigr\|_2^2. \tag{1}$$

Proposition 6.2 (KF loss is a leave-half-out estimate)**.** *Let C be drawn uniformly among the subsets of B of size $b/2$ and assume k restricted to $\{z_i\}_{i \in B}$ is strictly positive definite. Then $e_2(\theta; C) = \sum_{i \in B \setminus C} \|v_i - I_C(z_i)\|_2^2$, and*

$$\mathbb{E}_C\bigl[e_2(\theta; C)\bigr] \;=\; \frac{b}{2}\, \mathbb{E}_{C,\, i \sim \mathrm{Unif}(B \setminus C)}\bigl[\bigl\|v_i - I_C(z_i)\bigr\|_2^2\bigr],$$

i.e. $\frac{2}{b}\, e_2$ is an unbiased estimator, over the split randomness, of the expected squared error committed on a held-out point by the kernel predictor trained on a random half of the batch. When (B, C) are resampled at every step, the stochastic gradient $\nabla_\theta e_2(\theta; C)$ is unbiased for $\nabla_\theta\, \mathbb{E}_{B,C}[e_2(\theta; C)]$ whenever the expectation and derivative commute (e.g. under local Lipschitz domination).

The training loss we minimize for a neural mean is a sum of an empirical risk term and, optionally, $\beta\, e_2$: the first term is a linear functional of the empirical measure of the batch and measures fit, while Proposition 6.2 shows the second measures the *generalization of a kernel method built on the learned features*. This is the sense in which a KF-regularized network is trained to be a good feature map for the kernel correction that follows.

6.3 Stacking on a held-out split is safe

Between the means and the correction sits one more fitted object: the convex weights of the stacked ensemble, chosen to minimize the empirical risk on the validation split. Two elementary facts justify this step. First, by convexity, a convex combination of predictors is never worse than the corresponding weighted average of their risks, so stacking cannot be hurt by a weak member beyond its weight. Second, the weights live on a low-dimensional simplex and are fitted on hundreds of samples, so the selection cannot meaningfully overfit; the following proposition quantifies this with explicit constants.

Proposition 6.3 (validation stacking)**.** *Let $f_1, \dots, f_M : \mathcal{U} \to \mathcal{V}$ be fixed maps, let $\Delta = \{w \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1\}$, and for $w \in \Delta$ write $F_w = \sum_m w_m f_m$. Assume the members' per-sample relative errors are bounded: $\ell(f_m(u), G(u)) \leq B$ for μ-a.e. u and every m.*

(i) For every $w \in \Delta$, $\ell(F_w(u), G(u)) \le \sum_m w_m \,\ell(f_m(u), G(u))$ pointwise; in particular $R(F_w) \le \sum_m w_m R(f_m) \le \max_m R(f_m)$, and $\ell(F_w(u), G(u)) \le B$.

(ii) Let $\widehat{w}$ minimize the empirical risk $\widehat{R}(w) = \frac{1}{m}\sum_{i=1}^m \ell(F_w(\tilde{u}_i), G(\tilde{u}_i))$ over Δ, computed on an i.i.d. validation sample $\tilde{u}_1, \dots, \tilde{u}_m \sim \mu$ independent of $f_1, \dots, f_M$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$R(F_{\widehat{w}}) \;\le\; \min_{w \in \Delta} R(F_w) \;+\; \frac{8B}{m} \;+\; B\sqrt{\frac{2\big((M-1)\log\big(m(M-1)+1\big) + \log(2/\delta)\big)}{m}}.$$

With $M = 5$ members, $m = 1000$ validation samples and $\delta = 0.05$ the right-hand excess evaluates to about $0.28\,B$; at $B = 0.25$, the largest per-sample member error we observe, this caps the selection cost at roughly seven percentage points. The bound is conservative, as such bounds are: the realized validation-to-test gap of the stack in Section 4 is two orders of magnitude smaller. Its role is the scaling $\sqrt{M \log m / m}$, which says that fitting a handful of convex weights on a thousand held-out samples is statistically almost free; this is why the pipeline spends its validation data there and on the kernel hyperparameters and nowhere else.

The stack the pipeline deploys on the structural-mechanics benchmark is finer than a single convex vector: the weights are affine and vary per grid point, fitted by ridge on the validation split, and the choice between this object and the global convex stack is itself made on held-out data (fit both on one half of the split, keep whichever wins on the other half, refit the winner on the full split). Proposition 6.3 does not cover that construction; the following two statements do, in the same elementary style.

Lemma 6.4 (validated selection)**.** *Let $g_1, \dots, g_K : \mathcal{U} \to \mathcal{V}$ be fixed maps with $\ell(g_k(u), G(u)) \le B$ for μ-a.e. u and every k, and let $\widehat{k}$ minimize the empirical risk over an i.i.d. sample of size m' drawn from μ independently of everything used to build the g_k. Then, with probability at least $1 - \delta$,*

$$R(g_{\widehat{k}}) \;\le\; \min_{k \le K} R(g_k) \;+\; B\sqrt{\frac{2\log(2K/\delta)}{m'}}.$$

Here $K = 2$ and $m' = 500$: comparing two candidate stacks on half the validation split costs at most a few points of risk in the worst case, and in practice far less. The refit winner is covered by the next statement, which prices the per-coordinate affine fit itself.

Proposition 6.5 (per-pixel affine stacking)**.** *Fix members $f_1, \dots, f_M$, write $D = \sqrt{M+1}$, and for each output coordinate j let $\varphi_j(u) = (f_1(u)_j, \dots, f_M(u)_j, b) \in \mathbb{R}^{M+1}$, where b bounds all member and target values: $|G(u)_j| \le b$, $|f_m(u)_j| \le b$ for μ-a.e. u and all j, m. Consider the per-coordinate affine class $\{u \mapsto w^\top \varphi_j(u) : \|w\|_2 \le W\}$ and the coordinate risks $R_j(w) = \mathbb{E}\,(w^\top \varphi_j(u) - G(u)_j)^2$, estimated on an i.i.d. validation sample of size m independent of the members.*

(i) If $\widehat{w}_j$ minimizes the empirical risk of coordinate j over the class, then with probability at least $1 - \delta$,

$$\frac{1}{q}\sum_{j=1}^{q}\Big[R_j(\widehat{w}_j) - \min_{\|w\|_2 \le W} R_j(w)\Big] \;\le\; \frac{8\,Wb^2D\,(1+WD) + 2\,b^2(1+WD)^2\sqrt{\frac{1}{2}\log(4q/\delta)}}{\sqrt{m}}.$$

(ii) If instead $\widehat{w}_j$ minimizes the ridge objective $\widehat{R}_j(w) + \lambda\|w\|_2^2$ and satisfies $\|\widehat{w}_j\|_2 \le W$, the same bound holds with an additional λW^2 on the right.

The constants are crude, as in Proposition 6.3, and the reading is the same: with a handful of members the per-coordinate fit is paid for by the validation split at rate $\sqrt{(M+1)/m}$ per coordinate, uniformly over coordinates, and the ridge form the pipeline uses adds only its explicit penalty ($\lambda = 10^{-3}$ here, so λW^2 is negligible at the fitted weight norms, which the released runs report). What the bound buys is the license to fit $q(M+1)$ numbers on a thousand held-out samples without fearing selection bias at the reported precision; the honesty comparison of Lemma 6.4 then guards the one discrete choice that remains.

One stage of the pipeline is still uncovered at this point: the residual correction, whose scale and nugget are chosen from a finite grid on the same validation split that fitted the stack. The next statement closes the chain. Its key observation is structural: for a fixed grid point the correction is fit to the training residual of the stack, which is affine in the stack weights, so the corrected predictor is again a per-coordinate affine function of w with a transformed feature map, and the machinery of Proposition 6.5 applies to the composite class at the cost of one further measurable constant.

Corollary 6.6 (certificate for the deployed corrected surrogate)**.** *Assume the setting of Proposition 6.5, and let Γ be a finite set of correction configurations, each defining weights $c_\gamma(u) \in \mathbb{R}^{n_{\mathrm{tr}}}$ with the corrected predictor*

$$h_{w,\gamma}(u)_j \;=\; w_j^\top \varphi_j(u) + c_\gamma(u)^\top \big(Y_j - \Phi_j w_j\big),$$

where $Y_j \in \mathbb{R}^{n_{\mathrm{tr}}}$ collects the training targets of coordinate j and Φ_j the members' training predictions; $\gamma = 0$ (no correction, $c_0 \equiv 0$) is included in Γ. Suppose the correction weights are uniformly bounded, $\sup_{u,\gamma} \|c_\gamma(u)\|_1 \le \kappa$. If the deployed surrogate uses any validation-measurable choice $(\widehat{w}, \widehat{\gamma})$ with $\|\widehat{w}_j\|_2 \le W$ and $\widehat{\gamma}$ minimizing the empirical validation risk, aggregated over coordinates, among $\gamma \in \Gamma$ at the fitted $\widehat{w}$ (the grid choice is shared across coordinates, as in the pipeline), then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\widehat{\gamma}}\big) \;\le\; \min_{\gamma\in\Gamma}\; \frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\gamma}\big) + \frac{2(1+\kappa)^2\, b^2 (1+WD)\Big(4WD + (1+WD)\sqrt{\tfrac{1}{2}\log\big(4q|\Gamma|/\delta\big)}\Big)}{\sqrt{m}}.$$

Since $\gamma = 0$ is among the candidates, the oracle on the right is at most the fitted stack's own risk, and combining with Proposition 6.5(ii) bounds the deployed risk against the best affine stack itself. All three constants b, W and κ are measured and released with the runs.

The reading: the object the pipeline actually ships – fitted stack, grid-tuned correction, validated final choice – carries a finite-sample certificate whose every constant is observable, and the certificate is evaluated numerically alongside the seed experiments rather than left as a formula.

6.4 What the stack can achieve

Proposition 6.3 says fitting the weights is safe; it does not say mixing is worth anything. That is decided by a single measurable matrix. For a map f write the *normalized residual* at u as $\rho_f(u) = \big(f(u) - G(u)\big)/\|G(u)\|_2$, so that $\ell(f(u), G(u)) = \|\rho_f(u)\|_2$.

Proposition 6.7 (second-moment identity for convex stacks)**.** *Let $f_1, \dots, f_M$ be fixed maps and define $S \in \mathbb{R}^{M\times M}$ by $S_{mk} = \mathbb{E}_{u\sim\mu} \langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle$. For w in the simplex Δ^{M-1} let $f_w = \sum_m w_m f_m$. Then*

$$\mathbb{E}_{u\sim\mu}\, \ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S\, w,$$

so the squared-metric risk of every convex stack is determined by S, and the best achievable is $\min_{w\in\Delta^{M-1}} w^\top S w$. In particular:

(i) (two members) if S has diagonal (e_1^2, e_2^2) with $e_1 \le e_2$ and correlation $\varrho = S_{12}/(e_1e_2)$, mixing in the weaker member strictly helps if and only if $\varrho < e_1/e_2$, in which case the optimum is interior with value $e_1^2e_2^2(1-\varrho^2)/(e_1^2+e_2^2-2\varrho e_1e_2)$;

(ii) (equicorrelated family) if $S_{mm} = e^2$ and $S_{mk} = \varrho\, e^2$ for all $m \ne k$ with $\varrho \ge 0$, then $\min_w w^\top S w = e^2\big(\varrho + (1-\varrho)/M\big)$, attained at uniform weights and decreasing to $e^2\varrho$ as $M \to \infty$.

Part (i) is the classical two-member minimum-variance rule, and part (ii) is the equicorrelated case of the ambiguity decomposition of ensemble learning [12]; see Ueda and Nakano [26] for the generalization-error form and Brown et al. [4] for a survey. Both enter here as measurable decision tools on the estimated S.

Everything in this subsection is stated for the squared metric, and the tables of Sections 4 and 5 report a different one. The distinction is small numerically and decisive logically, so we make it explicit once and refer back to it wherever the two families of numbers meet.

Remark 6.8 (the two error metrics)**.** *For a map f write $\mathcal{E}_1(f) = \mathbb{E}\|\rho_f(u)\|_2$ and $\mathcal{E}_2(f) = (\mathbb{E}\|\rho_f(u)\|_2^2)^{1/2}$, the mean and the root mean square of the per-sample relative error. The tables report $\mathcal{E}_1$; Proposition 6.7 is a statement about $\mathcal{E}_2$, since $\mathcal{E}_2(f_w)^2 = w^\top S w$ by definition. Writing $X = \|\rho_f(u)\|_2$ and $c_f = (\operatorname{Var} X)^{1/2}/\mathbb{E}X$,*

$$\mathcal{E}_2(f)^2 - \mathcal{E}_1(f)^2 = \operatorname{Var} X, \qquad \mathcal{E}_1(f)/\mathcal{E}_2(f) = (1+c_f^2)^{-1/2} \le 1,$$

with equality only if the per-sample error is almost surely constant. Two consequences are used in this paper. The floor belongs in $\mathcal{E}_2$, because only $\mathcal{E}_2$ is determined by S: the map $w \mapsto w^\top S w$ is a quadratic form in second moments, whereas $w \mapsto \mathbb{E}\|\sum_m w_m \rho_{f_m}(u)\|_2$ depends on the full joint law of the residuals. And upper bounds transfer downward for free, since $\mathcal{E}_1 \le \mathcal{E}_2$, while lower bounds do not — so a floor read off S says nothing about a table entry without the dispersion of the same predictor. Comparing the two directly is biased in favour of the table entry by $(1+c_f^2)^{-1/2}$, which is about six percent on the structural-mechanics stack and ten to eleven percent on the O2 objects of Section 5; the dispersion is a property of the predictor and not a constant of the problem, so a single conversion factor should not be reused across objects.

Part (i) is an exact criterion, but it is used as a decision rule, and a decision rule is applied to estimates. Both ϱ and e_1/e_2 must be formed on data the decision does not see, so what governs the rule in practice is not the criterion but the gap it has to resolve relative to the noise in that estimation. The following makes this quantitative, and is stated for any threshold comparison of this shape.

Proposition 6.9 (margin law for a threshold rule)**.** *Let $D = \varrho - e_1/e_2$ be the signed gap of Proposition 6.7(i), so that mixing the weaker member strictly helps exactly when $D < 0$, and suppose $D \ne 0$. Let $\widehat{D}$ be an estimate of D formed on a split independent of the one on which D*

is evaluated, and suppose the estimation error $\widehat{D} - D$ is subgaussian with variance proxy σ^2, that is $\mathbb{E}\exp\big(\lambda(\widehat{D} - D)\big) \le \exp(\lambda^2\sigma^2/2)$ for all $\lambda \in \mathbb{R}$, conditionally on D. Then, writing $\Delta = |D|$ for the margin,

(i) on $\{D \neq 0\}$, the decision taken from $\widehat{D}$ differs from the correct one with probability at most $\exp(-\Delta^2/2\sigma^2)$;

(ii) for every $\tau \ge 0$, and with no condition on D,

$$\mathbb{P}\big(\operatorname{sign}\widehat{D} \neq \operatorname{sign} D, \quad |\widehat{D}| \ge \tau\big) \;\le\; \exp\bigg(-\frac{\tau^2}{2\sigma^2}\bigg).$$

Proof. Suppose first $D < 0$, so $\Delta = -D$ and the correct decision is to mix. The decision taken from $\widehat{D}$ differs only if $\widehat{D} \ge 0$, which forces $\widehat{D} - D \ge -D = \Delta$. The subgaussian hypothesis and the Chernoff bound give, for every $\lambda > 0$, $\mathbb{P}(\widehat{D} - D \ge \Delta) \le \exp(\lambda^2\sigma^2/2 - \lambda\Delta)$, and optimizing at $\lambda = \Delta/\sigma^2$ yields $\exp(-\Delta^2/2\sigma^2)$. If instead $D > 0$ then $\Delta = D$, the decision differs only if $\widehat{D} \le 0$, which forces $\widehat{D} - D \le -\Delta$, and the same argument applied to $-(\widehat{D} - D)$, which is subgaussian with the same proxy, gives the identical bound. The two cases are exhaustive because $D \neq 0$, which proves (i).

For (ii), condition on D again. If $D < 0$ the error event is $\{\widehat{D} \ge 0\}$, so on the error event intersected with $\{|\widehat{D}| \ge \tau\}$ we have $\widehat{D} \ge \tau$ and therefore $\widehat{D} - D \ge \tau - D \ge \tau$; the one-sided estimate above applies with Δ replaced by τ. If $D \ge 0$ the error event is $\{\widehat{D} < 0\}$, so on it with $|\widehat{D}| \ge \tau$ we have $\widehat{D} \le -\tau$ and therefore $D - \widehat{D} \ge \tau + D \ge \tau$, and the one-sided estimate in the other direction applies. In both cases the bound is $\exp(-\tau^2/2\sigma^2)$, which is free of D, so taking the expectation over D removes the conditioning. ☐

Three consequences are worth stating because they are what the measurement in Section 5.2 rests on. Part (ii) is the form the rule is actually used in: the quantity available when the decision is taken is $|\widehat{D}|$ and not Δ, so binning verdicts by the observed margin is legitimate, and it carries the same constant as (i). The bound is vacuous for $\Delta \lesssim \sigma$ and falls off quickly beyond it, so a rule of this shape has a band around its own threshold inside which it carries no guarantee, and the band is set by σ rather than chosen. Second, σ is estimable without knowing D: the differences between $\widehat{D}$ and the value of D realized on the held-out block, taken across many pairs, have exactly this scale. Third, and this is what makes the statement useful rather than merely cautionary, Δ is observable at decision time, so a practitioner can report the margin next to the verdict and know which verdicts the data support.

Exact equicorrelation never holds in practice, but for convex weights the floor survives one-sided bounds on the entries of S, which are what one actually measures.

Corollary 6.10 (ensembling floor)**.** *If every member satisfies $S_{mm} \ge \bar{e}^{\,2}$ and every pair satisfies $S_{mk} \ge \bar{\varrho}\,\bar{e}^{\,2}$ with $\bar{\varrho} \in [0,1]$, then every convex combination f_w obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S w \;\ge\; \bar{e}^{\,2}\big(\bar{\varrho} + (1-\bar{\varrho})\,\|w\|_2^2\big) \;\ge\; \bar{e}^{\,2}\,\bar{\varrho}.$$

No number of additional members with the same error level and mutual correlations moves the ensemble's root-mean-square relative error below $\bar{e}\sqrt{\bar{\varrho}}$.

The floor is a lower bound; a matching upper bound follows from the same entrywise information, and together they pin the achievable window from both sides by measured quantities alone.

Corollary 6.11 (the floor is nearly achieved)**.** *Suppose, in addition to the hypotheses of Corollary 6.10, that $S_{mm} \le \bar{e}^{\,2}(1+\delta_e)$ for every m and $S_{mk} \le (\bar{\varrho}+\delta_\varrho)\,\bar{e}^{\,2}$ for every $m \neq k$. Then*

$$\bar{\varrho} \;\le\; \frac{1}{\bar{e}^{\,2}} \min_{w \in \Delta^{M-1}} w^\top S w \;\le\; \bar{\varrho} + \delta_\varrho + \frac{1+\delta_e-\bar{\varrho}-\delta_\varrho}{M}.$$

The width of the bracket is the measured correlation spread δ_ϱ plus a $1/M$ term: when the members' errors and pairwise correlations are tight, whatever convex ensemble one builds sits within that window of the floor, and conversely no tuning of the weights can be blamed for more than the window.

On the structural-mechanics benchmark the measured spreads ($\varrho \in [0.83, 0.97]$ across member pairs and seeds, at member errors within a few percent of one another) put the window at roughly a tenth of the squared floor, and the fitted convex stack of Section 4 lands well inside it, within a percent of $\bar{e}\sqrt{\bar{\varrho}}$ in the squared metric. The bracket is evaluated against the realized ensemble risk at every seed, and its stability is the point: the predicted risk is $4.940 \pm 0.057\%$ and the realized risk a factor 0.938 ± 0.003 of it, so ten independent redraws of the validation split and of every network initialization move the prediction by six thousandths of a point and the ratio by three thousandths.

A second consequence of the same decomposition explains the final OCO-2 surrogate. Evaluation metrics there are diagonal in the output coordinates (the radiance metric is the weighting s_z), and diagonal metrics decouple.

Proposition 6.12 (per-coordinate combination under diagonal metrics)**.** *For weights $v = (v_{mj})$ that may depend on the output coordinate, let $f_v(u)_j = \sum_m v_{mj} f_m(u)_j$. For any diagonal metric with positive weights w,*

$$\mathbb{E}\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2 = \sum_j w_j^2\, \mathbb{E}\big(f_v(u)_j - G(u)_j\big)^2,$$

so the minimizing weights solve q separate problems, one per output coordinate, none of which involves w. The per-coordinate optimum is the same for every diagonal metric, and it weakly dominates every combination whose weights are shared across coordinates, for all such metrics simultaneously.

The practical content: when a problem is scored in several diagonal metrics at once, the combination stage does not need to know which one matters. Fit the best combination coordinate by coordinate on validation and the result serves all of them; only the members need metric-aware training, which is where the weighted mean of Section 5 enters.

The proposition as stated covers quadratic metrics with fixed diagonal weights, while the experiments report relative errors, which weight each sample by its output norm. The decoupling survives that weighting.

Corollary 6.13 (weighted metrics and the reported error)**.** *Let $a : \mathcal{U} \to [0,\infty)$ be measurable with $\mathbb{E}[a(u)\,\|f_v(u) - G(u)\|_2^2] < \infty$. Then, for every diagonal w,*

$$\mathbb{E}\Big[a(u)\,\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2\Big] = \sum_j w_j^2\, \mathbb{E}\Big[a(u)\,\big(f_v(u)_j - G(u)_j\big)^2\Big],$$

so the minimization still splits into q per-coordinate problems, none involving w. Taking $a(u) = \|G(u)\|_2^{-2}$ and $w = \mathbf{1}$ makes the left side the mean squared relative error: the combination *fitted coordinate by coordinate with per-sample weights a on validation is simultaneously optimal, within its class, for the squared relative error and for every diagonal reweighting of it. The reported metric, $\mathbb{E}\,\|\rho_{f_v}\|_2$ with ρ the normalized residual of Section 6.4, is controlled through $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$; the gap between the two sides is the dispersion factor whose measured value stays near* 0.93 *across the pipelines of this paper.*

The experiments run the combination both ways, unweighted and with the per-sample weights of the corollary, and report both.

Proposition 6.12 and its corollary describe the population optimum; the pipeline computes an empirical selection, coordinate by coordinate, on the validation split. The finite-sample cost of that step is priced by the same argument that priced the stacks.

Proposition 6.14 (empirical per-coordinate selection)**.** *Let $f_1, \ldots, f_M$ be fixed maps, $a : \mathcal{U} \to [0, A]$ a bounded per-sample weight, and suppose $a(u)\,(f_m(u)_j - G(u)_j)^2 \le L$ for μ-a.e. u, every member m and coordinate j. On a validation sample of size m_v independent of the members, let $\widehat{m}(j)$ minimize the empirical weighted risk of coordinate j over the M members, and let the combination take coordinate j from member $\widehat{m}(j)$. Then, with probability at least $1-\delta$, simultaneously for every coordinate,*

$$R_j\big(\widehat{m}(j)\big) \;\le\; \min_{m \le M} R_j(m) \;+\; 2L\sqrt{\frac{\log(2Mq/\delta)}{2m_v}}, \qquad R_j(m) = \mathbb{E}\big[a(u)\big(f_m(u)_j - G(u)_j\big)^2\big],$$

and consequently, for every diagonal metric w at once, $\sum_j w_j^2 R_j(\widehat{m}(j)) \le \sum_j w_j^2 \min_m R_j(m) + 2L\big(\sum_j w_j^2\big)\sqrt{\log(2Mq/\delta)/(2m_v)}$.

On the OCO-2 bands the deployed combination is exactly this estimator ($M = 6$ members, $q = 40$ coordinates, $m_v = 2000$), in the unweighted form and, for the squared relative error, with $a(u) = \|G(u)\|_2^{-2}$; the proposition prices both at the same rate, and the selection cost it bounds is what the seed experiments measure directly.

The corollary is the quantity this paper keeps measuring, at two different levels. Across architecture families on the structural-mechanics benchmark the pairwise correlations exceed 0.86 at member errors near 4.7%, which places the floor at $4.922 \pm 0.056\%$ in the root-mean-square metric it is stated in; Section 4.6 makes the comparison with the deployed pipeline in that metric rather than across the two. Across random seeds of one architecture on the OCO-2 bands the correlations are 0.78, 0.97 and 0.96 at member errors of 4.4%, 16.4% and 8.2%; measured in the squared metric of the proposition itself, over all 45 seed pairs per band, the floors are 4.360%, 17.970% and 9.248% and the realized ten-seed ensembles sit at 4.419%, 17.997% and 9.267% (Section 5). When an ensemble is at its floor the corollary says where improvement cannot come from. The reported tables use the mean rather than the root mean square of $\|\rho_f\|_2$, and the two are not interchangeable: Remark 6.8 gives the relation, the ratio runs from about 0.89 to 0.94 across the objects here, and every comparison in this paper between a quantity read off S and a table entry is made after converting one to the other. The proposition frames the diversity experiment of Section 4.6: what a new member is worth is legible in the row it adds to S, before any weights are fitted. On this benchmark the rows all look alike, pairwise correlations of 0.83–0.97 among trained networks of three architecture families and two losses, 0.86–0.88 against the kernel predictor, so the floor sits just below the best single member, and the measured stack lands on it. Part (i) is the same computation specialized to two members; it correctly predicts, from measured (e_1, e_2, ϱ) alone, which members receive weight in the fitted stack and which are dropped. Section 5 applies the same test on a problem where the answer comes out the other way: there the kernel's error is inflated by a factor the next subsection quantifies, the threshold e_1/e_2 collapses, and the kernel is dropped even at a residual correlation of 0.14.

6.5 Metric and representation: when a fixed kernel loses to a network

The structural-mechanics benchmark has the neural mean and the isotropic kernel within half a point of each other. On the OCO-2 emulation problem of Section 5 the same fixed kernel is an order of magnitude behind the same network. Two mechanisms can produce such a gap, and they separate cleanly.

Suppose the operator factors as $G(u) = g(Au)$ with $A \in \mathbb{R}^{d\times d}$ nonsingular and g of unit H^s norm, and that μ has a density bounded above and below on a compact domain. Order the singular values $\sigma_1 \geq \cdots \geq \sigma_d$ of A; they are the rates at which G varies along the input directions. Suppose A has *approximate rank* r at the sample size, meaning the trailing singular values fall below the achievable resolution, $\sigma_{r+1} \lesssim n^{-1/r}$, so the image $A\mathcal{U}$ is effectively r-dimensional at scale $n^{-1/r}$. The idealization beneath the display also takes the leading-r projection of the image design to keep a density bounded above and below, the trailing directions to be exact zeros at the working resolution, and the interpolation limit $\lambda \to 0$; Appendix A records the argument at that level, as a sketch rather than a complete proof.

Proposition 6.15 (isotropic and adapted kernel rates)**.** *Let $\widehat{G}_n^{\mathrm{iso}}$ be kernel ridge regression with a Matérn kernel of smoothness s and a validation-optimal single length scale, and $\widehat{G}_n^{\mathrm{ard}}$ the same construction in the metric $u \mapsto Au$. Up to constants depending on (σ_i), g, s and the density bounds, and up to logarithmic factors in n, with high probability over n i.i.d. samples,*

$$\mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{iso}}\big\| \;\lesssim\; \sigma_1^{\,s}\, n^{-s/d}, \qquad \mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{ard}}\big\| \;\lesssim\; n^{-s/r},$$

so the ratio of the two bounds is of order $\sigma_1^{\,s}\, n^{\,s(1/r-1/d)}$. When A is close to full rank ($r \approx d$) the two rates coincide and only the constant σ_1^s separates them.

The sketch (Appendix A) is the scattered-data estimate $\|G - \widehat{G}_n\| \lesssim h_X^s \|G\|_{H^s}$ applied to the two designs: one length scale must fill all d directions, so $h_X \asymp n^{-1/d}$ and the norm carries σ_1^s; the adapted metric, when A has a spectral gap, confines the design to r resolved directions and improves the fill distance. Both displays are upper bounds under the stated idealization — the exact metric A, a fixed g, and the length-scale selection folded into the logarithmic factors — and we claim no matching lower bound; the proposition is used here to calibrate what a metric can and cannot buy, not as a sharp rate theorem. The proposition bounds the *metric* part of a network's advantage, the part an anisotropic kernel would recover, and it says this part is a rate gain only under an approximate-rank gap; without one it is the constant σ_1^s alone. The rest is *representational*: a stationary kernel of fixed smoothness reaches only its native space, at any metric, while the network adapts its features to the map. The two parts are separated experimentally by handing the kernel the anisotropic metric and seeing how much of the gap closes. On the OCO-2 bands of Section 5 the metric closes a factor of 1.1 to 1.5 out of a tenfold gap (Section 5.1); the remainder is representational, and its direct evidence is that the same kernel machinery applied to the network's learned features recovers the network's accuracy and slightly exceeds it.

The rate half of the proposition is tested rather than assumed, and the test is the more interesting for not going our way. Both kernel rows are refit at nested training sizes over a sixteenfold range, at ten seeds per band. The sensitivity map's approximate rank is 0.54 to 0.60 of the input dimension, a gap wide enough that the proposition would permit a rate gain; the measured difference between the two empirical exponents is nevertheless small on all three bands and consistent with zero on one. The bound is an upper bound with no matching lower bound, so this does not contradict it — the adapted rate is free to beat its own bound — but it does mean that on these problems the rank gap is not converted into a rate, and a reader should take from Proposition 6.15 only the part the measurement supports: an anisotropic metric is worth a constant. The controlled rank-gap target of the verification suite, where the construction is exactly the idealization the proposition assumes, does show the adapted exponent improving; the distance between that and the OCO-2 bands is the distance between the idealization and a real sensitivity map.

The representational term has a precise reading through the same optimal-recovery bound that Section 6.6 makes exact. That bound factors the error as $\|G - m\|_{\mathcal{H}} \cdot P_\lambda$, a product of the target's norm in the kernel's native space and a design factor set by the Gram spectrum. Changing the features changes both in principle, but on the O2 band only one moves. The effective dimension of the Matérn Gram, the quantity Lemma 6.20 controls, is nearly identical on the raw input and on the learned features (at $n = 4000$ and a matched median length scale, 3995 against 3994 at a 10^{-8} nugget, with equal leading eigenvalue mass), so the design factor is essentially fixed. The native-space norm of the target is not: interpolating the same outputs through the two kernels, the raw-input kernel carries the target at squared norm 1.64×10^6 and the feature kernel at 3.87×10^4, a factor of 42. The network does not condition the kernel better; it moves the target to a low-norm corner of a native space of the same size, where the exact solve reaches it.

This is not an accident of the O2 band; it is what a pulled-back kernel does. Writing φ for the feature map and k for the base kernel, the deep kernel head uses $k_\varphi(x, x') = k(\varphi(x), \varphi(x'))$, the construction studied as deep kernel learning [5, 31].

Proposition 6.16 (feature pullback)**.** *The native space of k_φ is $\mathcal{H}_{k_\varphi} = \{f \circ \varphi : f \in \mathcal{H}_k\}$ with $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : f \circ \varphi = g \text{ on } \mathcal{X}\}$. In particular, if the target factors through the features as $G = h \circ \varphi$ with $h \in \mathcal{H}_k$, then $\|G\|_{\mathcal{H}_{k_\varphi}} \leq \|h\|_{\mathcal{H}_k}$, and the optimal-recovery bound of Theorem 6.17 for the feature-kernel regression is governed by $\|h\|_{\mathcal{H}_k}$ rather than by the norm of G in the raw-input space.*

The proof (Appendix A) is the standard pullback identity for reproducing kernels [23, 25]. Its content here is the reading: a fixed kernel on the raw input must carry the whole warped map G, whose native-space norm is large when G has fine structure; the same kernel on the features carries only the unwarped h, and training drives φ toward exactly the factorization that makes h simple. The measured factor of 42 is that norm gap, and it is the representational advantage that an anisotropic metric, which rescales the input but cannot re-express the map, leaves untouched (recovering only the factor 1.4).

6.6 An error bound for the residual kernel correction

The final stage of the pipeline is kernel ridge regression of the ensemble residual. The bound below is a vector-valued, residual form of the classical power-function estimate for kernel interpolation [30] and of the optimal-recovery viewpoint of Owhadi and Scovel [21]; we include the short argument in Appendix A to keep the two factors explicit. Fix the mean $m : \mathcal{U} \to \mathcal{V}$ (in the pipeline, the stacked ensemble; in the statement below m is any fixed map independent of the correction sample) and write $r = G - m$ for the residual operator with components r_j, $j = 1, \dots, q$. Let k be a positive definite kernel on $\mathcal{U}$ with RKHS $\mathcal{H}_k$, and assume $r_j \in \mathcal{H}_k$ for all j with

$$\|r\|_K^2 := \sum_{j=1}^{q} \|r_j\|_{\mathcal{H}_k}^2 < \infty.$$

Given inputs $X = (u_1, \dots, u_n)$, Gram matrix $K = k(X, X)$ and nugget $\lambda \geq 0$, the correction is $\widehat{r}_\lambda(u) = k(u, X)\,(K + n\lambda I)^{-1} R$, $R_{nj} = r_j(u_n)$, and the corrected predictor is $m + \widehat{r}_\lambda$. Let

$$P_\lambda(u)^2 \;=\; k(u, u) - k(u, X)\,(K + n\lambda I)^{-1} k(X, u) \tag{2}$$

denote the Gaussian process posterior variance with the same nugget (for $\lambda = 0$ this is the classical power function of the point set X).

Theorem 6.17 (certified pointwise bound)**.** *For every $u \in \mathcal{U}$ and every $\lambda \geq 0$,*

$$\big\|G(u) - m(u) - \widehat{r}_\lambda(u)\big\|_2 \;\leq\; \|G - m\|_K\, \widetilde{P}_\lambda(u) \;\leq\; \|G - m\|_K\, P_\lambda(u),$$

where $\widetilde{P}_\lambda(u)^2 = P_\lambda(u)^2 - n\lambda\, \|(K + n\lambda I)^{-1} k(X, u)\|_2^2$.

Three comments. First, the inequality is algebraic: it holds pathwise for every fixed m, every dataset, and every u, with no probabilistic assumptions. In the pipeline m is itself fit on the same training set; this does not affect the validity of the bound, only the reading of $\|G - m\|_K$ as a random (realized) quantity rather than an a priori one. Second, the bound factorizes into a quantity that depends only on the residual operator ($\|G - m\|_K$) and a quantity that depends only on the kernel and the design (P_λ), and the second factor is exactly the posterior standard deviation that a Gaussian process interpretation of the correction would report. This is the sense in which the bound is certified: whatever error the corrected surrogate commits at u is controlled by the reported $P_\lambda(u)$ times a constant that does not depend on u. We stress that this is a statement about a valid upper bound, not about the usefulness of P_λ as a pointwise error *ranking*. Section 4.8 finds that on this benchmark P_λ ranks the error poorly, because the neural mean supplies most of the error and the relative metric is confounded by output amplitude; a distribution-free conformal rescaling of P_λ nonetheless attains its nominal coverage on the test set. Third, the theorem quantifies why one should regress residuals rather than raw targets: the design factor P_λ is the same in both cases, so the gain of the neural mean is the drop from $\|G - \bar{v}\|_K$ (kernel-only, mean $\bar{v}$) to $\|G - m\|_K$. These norms are not directly observable, but the RKHS norms of the *fitted* interpolants, $\|\widehat{r}_\lambda\|_K^2 = \operatorname{tr}(\alpha^\top K \alpha)$ with $\alpha = (K + n\lambda I)^{-1} R$, are computable lower-bound proxies, and they drop by a factor of about 271 when the kernel is moved from raw targets to ensemble residuals (Table 6). The neural mean does not merely reduce the size of the residual in L^2; it leaves behind a residual operator that is genuinely smoother as seen by the kernel.

6.7 Distribution-free coverage for the reported band

Theorem 6.17 controls the error by $\|G - m\|_K\, P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori, and Section 4.8 shows that P_λ alone ranks the error weakly. What the surrogate reports as uncertainty is therefore not P_λ itself but a rescaling of it, $q\, P_\lambda(u)$, whose multiplier q is calibrated on the held-out split by the split-conformal rule

$$q = Q_{1-\alpha}\Big(\big\{s_i := \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)\big\}_{i=1}^{m}\Big), \qquad e(u) = G(u) - m(u) - \widehat{r}_\lambda(u), \tag{3}$$

where $Q_{1-\alpha}$ is the $\lceil(1-\alpha)(m+1)\rceil$-th smallest value of the m validation scores. The point of this construction is that its coverage needs neither the bound to be tight nor P_λ to be a good error ranking; it needs only exchangeability, which the fixed splits provide.

Proposition 6.18 (finite-sample coverage)**.** *Fix the mean m, the correction $\widehat{r}_\lambda$ and the design X, and let $P_\lambda(\cdot) > 0$. Suppose the validation and test inputs $\tilde{u}_1, \dots, \tilde{u}_m, u^\star$ are exchangeable (in particular i.i.d. from μ) and drawn independently of $m, \widehat{r}_\lambda, X$. Let q be the conformal multiplier (3) and define the band $C(u) = \{v : \|v - m(u) - \widehat{r}_\lambda(u)\|_2 \le q\, P_\lambda(u)\}$. Then*

$$1-\alpha \;\le\; \mathbb{P}\big(G(u^\star) \in C(u^\star)\big) \;\le\; 1-\alpha+\frac{1}{m+1},$$

the upper bound holding when the scores are almost surely distinct.

Remark 6.19 (the exact law of realized coverage)**.** *When the calibration and test scores are i.i.d. with a continuous distribution F — as in the campaign's protocol, where calibration and evaluation sets are a random split of an i.i.d. test block — the coverage probability conditional on the calibration sample is F evaluated at the k-th smallest calibration score, $k = \lceil(1-\alpha)(m+1)\rceil$. Since $F(s_i)$ are i.i.d. uniform, this is the k-th order statistic of m uniforms, with distribution exactly* $\mathrm{Beta}(k,\, m+1-k)$ *[28]; its mean $k/(m+1)$ recovers Proposition 6.18. Conditional on the calibration sample the evaluation indicators are i.i.d. Bernoulli, so the coverage observed on N evaluation points follows the Beta–Binomial law with these parameters. Each seed's observed coverage is therefore placed against the central band of an exact finite-sample law, whose per-seed width at $m = 1000$ is close to $\sqrt{\alpha(1-\alpha)/m}$.*

The two-sided statement is the standard guarantee of split conformal prediction [14, 29], transported to the functional-output setting by taking the nonconformity score to be the relative residual norm $s = \|e(u)\|_2/P_\lambda(u)$; the proof, a rank argument on the exchangeable scores, is in Appendix A. Three points make it the right closing statement for the uncertainty analysis. It is close to the quantity we compute: the reported coverage of $90.2\pm0.6\%$ at the nominal $1-\alpha = 90\%$ in Section 4.8 realizes this proposition with $m = 1000$ and $\alpha = 0.1$. Any gap to nominal is not the $1/(m+1)$ slack, which is only a tenth of a point; it is the finite-sample fluctuation of conditional coverage for a single calibration draw, whose standard deviation $\sqrt{\alpha(1-\alpha)/m}$ is about a point at these sizes. Remark 6.19 gives the exact law of that fluctuation, and the ten seeds let it be checked rather than invoked: the observed spread across seeds is 0.6 points against the 0.9 points the law predicts, and every seed falls inside the central 95% band of the exact distribution at the 90% level and at the 95% level alike. The hypotheses hold exactly as stated: calibration uses a random subset of the test block that no training, checkpointing, stacking, or tuning step ever touched, and coverage is evaluated on the disjoint remainder (Appendix B), so the exchangeability the proposition assumes is exact, with no low-complexity selection argument needed. It is agnostic to everything the earlier results leave uncertain: the neural mean may be biased, P_λ may correlate with the error weakly or with the wrong sign, and the bound of Theorem 6.17 may be loose, yet the band still covers at the stated rate. And it is where the uncertainty analysis ends: among the candidate uncertainty signals we examine, the conformally rescaled P_λ is the only one that carries a guarantee, so it is the one the surrogate reports.

6.8 Effective dimension from the Gram spectrum

The remaining design choice is the nugget λ, selected by cross-validation in the pipeline. The following exact identity connects that choice to the spectrum of the Gram matrix and, through it, to random matrix descriptions of the design.

Lemma 6.20 (effective dimension and the empirical Stieltjes transform)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$, let $\widehat{m}(z) = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i/n - z)^{-1}$ be the Stieltjes transform of the empirical spectral distribution of K/n. Then for every $\lambda > 0$ the effective dimension of ridge regression with nugget λ satisfies*

$$d_{\text{eff}}(\lambda) \;:=\; \operatorname{tr}\big(K(K + n\lambda I)^{-1}\big) \;=\; n\big(1 - \lambda\, \widehat{m}(-\lambda)\big).$$

In particular, if the empirical spectral distribution of K/n converges weakly to a limit law F as $n \to \infty$, then $d_{\text{eff}}(\lambda)/n \to 1 - \lambda\, m_F(-\lambda)$ with m_F the Stieltjes transform of F.

The identity is unconditional; the random-matrix content enters through the choice of F. For the benchmark's simplest reference model, isotropic linear features, the limit can be evaluated in closed form.

Corollary 6.21 (effective dimension under a Marčenko–Pastur limit)**.** *Let $K = XX^\top$ with $X \in \mathbb{R}^{n\times p}$ having i.i.d. entries of mean zero and variance σ^2, and let $p/n \to \gamma \in (0, 1]$. Then*

$$\frac{d_{\text{eff}}(\lambda)}{n} \;\longrightarrow\; \gamma\big(1 - \lambda\, m(-\lambda)\big), \qquad m(-\lambda) = \frac{\sqrt{\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda} - \big(\lambda + \sigma^2(1-\gamma)\big)}{2\gamma\sigma^2\lambda},$$

almost surely, where m is the Stieltjes transform of the Marčenko–Pastur law with ratio γ and scale σ^2. In a simulation with $n = 3000$, $p = 900$, $\sigma^2 = 1.7$, the formula agrees with the empirical $d_{\text{eff}}(\lambda)/n$ to four decimal places across $\lambda \in [10^{-3}, 10]$.

Two uses. Quantitatively, the formula makes the nugget–capacity trade-off explicit for the bulk of a feature Gram matrix: eigenvalue mass of Marčenko–Pastur type is absorbed or discarded by λ according to a single closed-form curve, and outlying (spiked) eigenvalues x_s simply add their terms $x_s/(x_s + n\lambda)$ on top. Qualitatively, it separates the two regimes visible in Figure 7: the Matérn Gram matrix on the loads is nothing like a Marčenko–Pastur bulk (its spectrum decays exponentially, which is why d_{eff} is a few hundred out of 19000), whereas the Gram matrices of learned penultimate features do show a bulk-plus-spikes shape, and for them the corollary describes how much of the bulk the cross-validated nugget retains. For the feature Gram matrices of the trained networks we observe the familiar picture of a bulk that is well fitted by a Marčenko–Pastur law [19] together with a small number of outlying eigenvalues carrying the regression signal, as in spiked covariance models [2]; for the Matérn Gram matrix on the 41-dimensional loads the spectrum decays rapidly and d_{eff} is small ($\approx$ a few hundred at the cross-validated λ, out of $n = 19000$; Figure 7). This gives a post-hoc reading of the cross-validated nugget: λ lands where $d_{\text{eff}}(\lambda)$ has absorbed the outlying eigenvalues and the leading bulk, and further decreasing λ buys capacity precisely where the spectrum carries little signal. It also explains why the exact solve is affordable: the correction is numerically a problem of dimension d_{eff}, not n.

The effective dimension is also exactly the total budget of the design factor from Theorem 6.17. Writing $A = K + n\lambda I$, the posterior variances at the training inputs sum to

$$\sum_{i=1}^{n} P_\lambda(u_i)^2 = \operatorname{tr}(K) - \operatorname{tr}(KA^{-1}K) = \operatorname{tr}\big(n\lambda\, KA^{-1}\big) = n\lambda\, d_{\text{eff}}(\lambda),$$

using $KA^{-1}K = K - n\lambda\, KA^{-1}$. So the same d_{eff} that reads the nugget off the spectrum also fixes, up to the factor $n\lambda$, the total posterior-variance mass the correction distributes over the design; the two halves of this section are one quantity seen from two sides.

7 Discussion

What moved the number, and what stopped it. The improvement over the published band decomposes unevenly, and with ten replicates the pieces can be sized rather than asserted. Accurate neural means trained on the reported metric and averaged over the reflection symmetry do most of the work: from the kernel's 5.197% to the best single network's 4.712%. Combining members adds what their measured residual correlations permit and no more — 0.064 points from equal weighting, 0.021 from weights fitted on validation, 0.032 from letting those weights vary over the grid — and the kernel correction contributes 0.024 ± 0.005 points, bringing with it the certificate of Theorem 6.17, the end-to-end guarantee of Corollary 6.6, and the calibrated band of Proposition 6.18. Each of those contributions is smaller than the difference between two published architectures, and only the seeds make them separable from noise at all.

The largest single lever is none of them: it is which members the stack contains. Removing one architecture, the spectral one, costs 0.039 ± 0.011 points — more than any individual stage above — even though that member ranks only third of five on the metric. This is the practical form of the second-moment reading. What a member is worth to a stack is set by how its errors covary with the others', not by how small they are, and a stack of near-duplicates cannot be repaired by tuning the combination harder.

We did not find evidence that any published architecture was mis-tuned by its authors; the components compose, and the composition had not been tried on this benchmark. What stops further movement is not a missing architecture. Section 4.6 locates the remaining error in a component that all families share, that concentrates where the finite element data are least reliable, that no input statistic predicts, and that neither capacity, nor diversity, nor more data, nor reseeding reduces; Section 4.7 adds that the cases where it is worst are the cases where every member fails at once. The margin over the published methods other than PARA-Net is real and survives both error bars of Section 4.4. Against PARA-Net the five-member pipeline is level: 0.022 points, about one standard error, which we read as a tie and not as a win. Without the spectral member the same pipeline loses by 0.061 points, about three standard errors, so the tie belongs to the configuration that carries that member rather than to the approach in general. An earlier version of this work claimed a match, on a single run. All of these should be read for what they are: the last few learnable hundredths above what the evidence of Section 4.6 reads as a data-set floor, not a step on the way to zero.

Relation to neural-mean Gaussian processes. Our correction stage is closest in spirit to Mora et al. [20], who place a neural operator inside the mean of a Gaussian process and fit both jointly. The differences are operational but they matter at this scale: we fit the mean first and the kernel after (so the expensive stage is ordinary network training), we tune the kernel by validation rather than by marginal likelihood, we correct with an exact solve at $n = 19000$ rather than train through the kernel, and we add the feature-space second stage. The theory in Section 6 applies verbatim to their setting as well; the measured drop in the fitted RKHS norm (Table 6) is the quantitative reason residual corrections of accurate means are the right place to spend kernel capacity.

Relation to kernel emulation practice. The pipeline is consistent with the experience of the kernel-flow line of work in emulation at scale. The closest instance is the forward-model emulator of Lamminpää et al. [13] for the OCO-2 CO_2 retrievals, in which a Gaussian process with a cross-validation-learned kernel replaces the full-physics radiative transfer code within measurement-error precision; kernel flows have likewise been used to infer convective-storm structure from passive microwave observations [24]. The lesson of that work matches ours: kernels with data-adapted hyperparameters are extremely effective once the input is presented in its physical parametrization and the target has been reduced to something smooth. Here the reduction is performed by the neural ensemble rather than by physics, and the kernel-flow loss itself admits the leave-half-out reading of Proposition 6.2. Section 5 carries out the comparison against that emulator on its own test points; the step this paper does not take is the retrieval-level one. The mission's Level 1 and Level 2 products are publicly distributed through NASA's Earthdata archive, and evaluating how the surrogate's certified bands and fallback behavior propagate through an actual retrieval, rather than through the emulation metrics alone, is the natural continuation and requires the estimation machinery of the retrieval pipeline rather than new data collection.

Limitations. The benchmark has a one-dimensional input function and a fixed geometry; the exact kernel solve exploits both, and problems with high-dimensional input fields would require the usual approximations (inducing points, random features) at some cost to the certificates. The certified bound of Theorem 6.17 controls the error relative to the RKHS norm of the residual operator, a quantity we can only lower-bound empirically; the conformal calibration we report is the honest, assumption-light complement. Training the means on the metric, the reflection steps, and the stacking are benchmark-agnostic, but the specific error levels reported here should be read as properties of this dataset and protocol.

The seed coverage is uneven and it is uneven in the places that matter least for the paper's arguments but most for a reader who wants a single number. The structural-mechanics members, stack, correction, second-moment measurement and conformal calibration are replicated ten times. The six-member configuration that produced our lowest error completed at two of the ten seeds, its UNet member having been scheduled last; both runs land at 4.539% and 4.537%. Two seeds are not ten, so we report them and press nothing on them. The low-data pipeline now carries ten seeds, and its margin over the best published row is 1.06 ± 0.09 points; what remains single-seeded in Table 3 is the kernel-ridge baseline. The OCO-2 tables carry ten seeds per band, and the scaling sweep behind Section 5.1 agrees with them at the size where the two overlap. Our low-data protocol reproduces that of Mora et al. [20] up to the unavoidable ambiguity of which 1250 samples are used; we use the first 1250 of the training block and report the same 20000-sample test set, and we release splits and code so the comparison can be audited.

One prediction of the theory did not survive the extra measurement. Proposition 6.15 allows an anisotropic metric a rate gain when the sensitivity map has an approximate-rank gap, and the OCO-2 bands have one, at a rank ratio between 0.54 and 0.60. Refitting both kernel rows across a sixteenfold range of training sizes at ten seeds finds no rate gain on one band and a small one on the other two, well short of what the rank gap would license; what the metric delivers is a constant between 1.1 and 1.5 (Section 5.1). Since the proposition is an upper bound with no matching lower bound, this is not a contradiction, but it does mean the rate half of it earns no experimental support here, and we have restated the conclusion to the constant half, which the data do support. An earlier version of this work reported the O2 band's effective rank as near-full and read the absence of a rate gain as confirmation; the campaign's causal, per-seed measurement puts that rank at 12 of 20, which makes the same observation a failed prediction rather than a confirmed one.

What the second problem settles. The OCO-2 study keeps every component and inverts the balance, and the pair of problems brackets the design space. When the network and the kernel tie (structural mechanics), the coupling pays: stack them, correct the residual, and the certified and conformal machinery rides along. When the kernel trails by an order of magnitude (OCO-2), the coupling is not the point; the kernel's value moves inside the network, as an exact head on its learned features, and the pipeline's job becomes metric management, training each member in the metric it will be scored in and combining per coordinate. In both regimes the same two measurements decide everything in advance: the pairwise residual correlations, which set the ensembling floor of Corollary 6.10 at whatever level the members share their errors and, through Corollary 6.11, bracket how close the fitted ensemble must come to it, and the member error ratio, which the second-moment identity turns into a keep-or-drop verdict for each candidate. Nothing in that protocol is specific to these two datasets.

Outlook. Three directions seem worth pursuing. First, the benchmark itself: regenerating the dataset with refined meshes at the corner singularities, higher-order elements, or output grids that resolve the concentrations would move the floor that Section 4.6 measures, and would return the problem to being a test of surrogates rather than of its own data. The diagnostic protocol used there, cross-family residual correlation, the second-moment prediction of Proposition 6.7, and the scaling-in-n check, is cheap to run on any benchmark suspected of the same condition. Second, the same recipe on the remaining benchmarks of de Hoop et al. [8] and Batlle et al. [3], where the input functions are genuinely high-dimensional and the interplay between neural means and kernel corrections should look different. Third, the uncertainty side: P_λ is a design quantity, so it can be optimized, and the connection of Lemma 6.20 between the nugget, the spectrum, and the effective dimension suggests principled ways to spend a fixed computational budget on the correction stage.

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Declaration of competing interest

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. A large part of the software was written with Fable 5 (Anthropic): the implementation of the experiments and diagnostics, the downloading, assembly and preparation of the benchmark and OCO-2 datasets, and the adaptation of the publicly released emulator code of Lamminpää et al. — originally split between Julia and a ReFRACtor-driving Python layer — into a single Python pipeline for sampling states and reducing radiances. The main contributions are the author's: the residual coupling of neural means and exact kernel corrections, the two-regime framing that organizes the paper, and the supporting theory, all developed with AI assistance in calculation, drafting, and exposition. Each theorem and its proof was checked independently and more than once — by Fable 5, by ChatGPT Pro (OpenAI), and by the author — and the reported numbers were verified against the released per-run data. The author reviewed all of the above and takes full responsibility for the content of the manuscript.

Data and code availability

The code, the per-run summaries behind every reported number, and the exact configuration of each experiment are available at `https://github.com/yspennstate/neural-means-kernel-corrections`. The structural-mechanics, Helmholtz, Navier–Stokes and advection data are distributed through the data record accompanying de Hoop et al. [8] (`data.caltech.edu`, record `20091`); the OCO-2 emulation data and the kernel-flow emulator's predictions through the OSF project `u2t8a`; and the ClimSim data through the LEAP repository `subsampled_low_res`.

References

[1] R. A. Adams and J. J. F. Fournier. *Sobolev Spaces*. Academic Press, 2nd edition, 2003.

[2] J. Baik, G. Ben Arous, and S. Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Annals of Probability*, 33(5):1643–1697, 2005.

[3] P. Batlle, M. Darcy, B. Hosseini, and H. Owhadi. Kernel methods are competitive for operator learning. *Journal of Computational Physics*, 496:112549, 2024.

[4] G. Brown, J. Wyatt, R. Harris, and X. Yao. Diversity creation methods: a survey and categorisation. *Information Fusion*, 6(1):5–20, 2005.

[5] R. Calandra, J. Peters, C. E. Rasmussen, and M. P. Deisenroth. Manifold Gaussian processes for regression. In *International Joint Conference on Neural Networks (IJCNN)*, pages 3338–3345, 2016.

[6] A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.

[7] M. Darcy, B. Hamzi, G. Livieri, H. Owhadi, and P. Tavallali. One-shot learning of stochastic differential equations with data adapted kernels. *Physica D: Nonlinear Phenomena*, 444: 133583, 2023.

[8] M. V. de Hoop, D. Z. Huang, E. Qian, and A. M. Stuart. The cost-accuracy trade-off in operator learning with neural networks. *Journal of Machine Learning*, 1(3):299–341, 2022.

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[10] V. Koltchinskii and E. Giné. Random matrix approximation of spectra of integral operators. *Bernoulli*, 6(1):113–167, 2000.

[11] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[12] A. Krogh and J. Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems 7*, pages 231–238. MIT Press, 1995.

[13] O. Lamminpää, J. Susiluoto, J. Hobbs, J. McDuffie, A. Braverman, and H. Owhadi. Forward model emulator for atmospheric radiative transfer using Gaussian processes and cross validation. *Atmospheric Measurement Techniques*, 18:673–694, 2025.

[14] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523): 1094–1111, 2018.

[15] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[16] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[17] L. Lu, X. Meng, S. Cai, Z. Mao, S. Goswami, Z. Zhang, and G. E. Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*, 393:114778, 2022.

[18] C. Lyle, M. van der Wilk, M. Kwiatkowska, Y. Gal, and B. Bloem-Reddy. On the benefits of invariance in neural networks. *arXiv preprint arXiv:2005.00178*, 2020.

[19] V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

[20] C. Mora, A. Yousefpour, S. Hosseinmardi, H. Owhadi, and R. Bostanabad. Operator learning with Gaussian processes. *Computer Methods in Applied Mechanics and Engineering*, 434: 117581, 2025.

[21] H. Owhadi and C. Scovel. *Operator-Adapted Wavelets, Fast Solvers, and Numerical Homogenization.* Cambridge University Press, 2019.

[22] H. Owhadi and G. R. Yoo. Kernel flows: from learning kernels from data into the abyss. *Journal of Computational Physics*, 389:22–47, 2019.

[23] V. I. Paulsen and M. Raghupathi. *An Introduction to the Theory of Reproducing Kernel Hilbert Spaces.* Cambridge University Press, 2016.

[24] S. Prasanth, Z. S. Haddad, J. Susiluoto, A. J. Braverman, H. Owhadi, B. Hamzi, S. M. Hristova-Veleva, and J. Turk. Kernel flows to infer the structure of convective storms from satellite passive microwave observations. AGU Fall Meeting Abstracts, abstract A55F-1445, 2021.

[25] S. Saitoh. *Integral Transforms, Reproducing Kernels and their Applications*. Longman, 1997.

[26] N. Ueda and R. Nakano. Generalization error of ensemble estimators. In *Proceedings of the IEEE International Conference on Neural Networks*, volume 1, pages 90–95, 1996.

[27] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[28] V. Vovk. Conditional validity of inductive conformal predictors. In *Proceedings of the Asian Conference on Machine Learning*, volume 25 of *PMLR*, pages 475–490, 2012.

[29] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

[30] H. Wendland. *Scattered Data Approximation*. Cambridge University Press, 2004.

[31] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing. Deep kernel learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, volume 51 of *Proceedings of Machine Learning Research*, pages 370–378, 2016.

[32] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[33] G. R. Yoo and H. Owhadi. Deep regularization and direct training of the inner layers of neural networks with kernel flows. *Physica D: Nonlinear Phenomena*, 426:132952, 2021.

[34] S. Yu, W. M. Hannah, L. Peng, J. Lin, M. A. Bhouri, R. Gupta, et al. ClimSim: A large multi-scale dataset for hybrid physics-ML climate emulation. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36, 2023.

A Proofs

A.1 Proof of Proposition 6.1

By convexity of $\ell(\cdot, v)$,

$$\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \tfrac{1}{2}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\ell\big(T\widehat{G}(Su), G(u)\big).$$

Since T is an involution, the equivariance $G(Su) = TG(u)$ applied at u gives $TG(Su) = T^2G(u) = G(u)$, hence

$$\ell\big(T\widehat{G}(Su), G(u)\big) = \ell\big(T\widehat{G}(Su), TG(Su)\big) = \ell\big(\widehat{G}(Su), G(Su)\big),$$

using the T-invariance of the loss. Taking expectations and using the S-invariance of μ (so that $Su \sim \mu$ when $u \sim \mu$),

$$\mathbb{E}\,\ell\big(\widehat{G}_S(u), G(u)\big) \le \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(Su), G(Su)\big) = \mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big). \quad \square$$

A.2 Proof of Proposition 6.2

Strict positive definiteness of the restricted kernel matrix makes I_C the (unique) interpolant of the pairs indexed by C, so $I_C(z_j) = v_j$ for $j \in C$ and the terms of (1) indexed by C vanish, which is the first claim. For the second, write

$$e_2(\theta; C) \;=\; \sum_{i \in B \setminus C} g(C, i), \qquad g(C, i) := \|v_i - I_C(z_i)\|_2^2.$$

Conditionally on C, the sum has exactly $b/2$ terms, so $e_2(\theta; C) = \frac{b}{2}\,\mathbb{E}_{i \sim \mathrm{Unif}(B \setminus C)}[g(C, i)]$, and taking the expectation over C proves the identity. The gradient claim is immediate: for fixed (B, C) the map $\theta \mapsto e_2(\theta; C)$ is differentiable wherever the Cholesky factorization in I_C is (the kernel matrix stays positive definite in a neighborhood), and under a local integrable Lipschitz bound the derivative passes under the expectation, so $\mathbb{E}_{B,C}[\nabla_\theta e_2] = \nabla_\theta\, \mathbb{E}_{B,C}[e_2]$. □

A.3 Proof of Proposition 6.3

(i) Since $\sum_m w_m = 1$, $F_w(u) - G(u) = \sum_m w_m\,(f_m(u) - G(u))$, and the triangle inequality gives $\|F_w(u) - G(u)\|_2 \le \sum_m w_m \|f_m(u) - G(u)\|_2$. Dividing by $\|G(u)\|_2$ yields the pointwise claim; taking expectations gives the risk inequality, and bounding each $\ell(f_m(u), G(u))$ by B gives $\ell(F_w(u), G(u)) \le B$.

(ii) Write $\ell_u(w) = \ell(F_w(u), G(u))$. First, ℓ_u is Lipschitz on Δ for the ℓ^1 norm: for $w, w' \in \Delta$, the reverse triangle inequality and the argument of (i) give

$$|\ell_u(w) - \ell_u(w')| \;\le\; \frac{\big\|\sum_m (w_m - w'_m)(f_m(u) - G(u))\big\|_2}{\|G(u)\|_2} \;\le\; B\,\|w - w'\|_1.$$

Next, discretize the simplex. For $k \in \mathbb{N}$ let $\mathcal{G}_k = \{w \in \Delta : kw \in \mathbb{Z}^M\}$; its cardinality is the number of compositions of k into M nonnegative parts, $\binom{k+M-1}{M-1} \le (k+1)^{M-1}$. Given $w \in \Delta$, set $w'_i = \lfloor k w_i \rfloor / k$ for $i < M$ and $w'_M = 1 - \sum_{i<M} w'_i$; then $w' \in \mathcal{G}_k$ (the last coordinate is a multiple of $1/k$ and is $\ge w_M \ge 0$ because the first $M-1$ coordinates only decreased), and

$$\|w - w'\|_1 = \sum_{i<M} (w_i - w'_i) + \big(w'_M - w_M\big) = 2 \sum_{i<M} (w_i - w'_i) \;\le\; \frac{2(M-1)}{k}.$$

By (i) the per-sample loss lies in $[0, B]$, so for each fixed w' Hoeffding's inequality gives $\mathbb{P}\big(|\widehat{R}(w') - R(w')| > t\big) \le 2e^{-2mt^2/B^2}$, and a union bound over $\mathcal{G}_k$ shows that with probability at least $1 - \delta$,

$$\sup_{w' \in \mathcal{G}_k} \big|\widehat{R}(w') - R(w')\big| \;\le\; B\sqrt{\frac{\log\big(2|\mathcal{G}_k|/\delta\big)}{2m}}.$$

On this event, for every $w \in \Delta$, approximating by its grid point and using the Lipschitz property for both R and $\widehat{R}$,

$$\big|\widehat{R}(w) - R(w)\big| \;\le\; B\sqrt{\frac{\log(2|\mathcal{G}_k|/\delta)}{2m}} + \frac{4B(M-1)}{k}.$$

If $w^\star$ minimizes R over Δ, the empirical minimizer satisfies $R(F_{\widehat{w}}) \le \widehat{R}(\widehat{w}) + \sup_\Delta |\widehat{R} - R| \le \widehat{R}(w^\star) + \sup_\Delta |\widehat{R} - R| \le R(F_{w^\star}) + 2\sup_\Delta |\widehat{R} - R|$. Choosing $k = m(M-1)$ and using $|\mathcal{G}_k| \le$

$(m(M-1)+1)^{M-1}$ gives

$$R(F_{\widehat{w}}) \le \min_{w\in\Delta} R(F_w) + \frac{8B}{m} + B\sqrt{\frac{2\big((M-1)\log(m(M-1)+1)+\log(2/\delta)\big)}{m}},$$

which is the claim. $\square$

A.4 Proof of Lemma 6.4

For each k the validation scores $\ell(g_k(\tilde{u}_i), G(\tilde{u}_i))$, $i \le m'$, are i.i.d., take values in $[0, B]$, and have mean $R(g_k)$; the maps g_k are fixed relative to this sample, which is the only place the independence assumption enters. Hoeffding's inequality gives, for each k,

$$\Pr\Big(\big|\widehat{R}(g_k) - R(g_k)\big| \ge t\Big) \le 2\exp\big(-2m't^2/B^2\big),$$

so with $t = B\sqrt{\log(2K/\delta)/(2m')}$ and a union bound over $k \le K$, all K deviations are below t simultaneously with probability at least $1-\delta$. On that event, if k^* attains $\min_k R(g_k)$,

$$R(g_{\widehat{k}}) \le \widehat{R}(g_{\widehat{k}}) + t \le \widehat{R}(g_{k^*}) + t \le R(g_{k^*}) + 2t,$$

and $2t = B\sqrt{2\log(2K/\delta)/m'}$. $\square$

A.5 Proof of Proposition 6.5

Fix a coordinate j and abbreviate $\varphi = \varphi_j$, $Y(u) = G(u)_j$, $D = \sqrt{M+1}$. Each entry of $\varphi(u)$ is bounded by b in absolute value (the members by assumption, the intercept entry by construction), so $\|\varphi(u)\|_2 \le bD$, and for $\|w\|_2 \le W$ Cauchy–Schwarz gives $|w^\top\varphi(u)| \le WbD$ and $|w^\top\varphi(u) - Y(u)| \le b(1+WD)$. The losses $\ell_w(u) = (w^\top\varphi(u) - Y(u))^2$ therefore take values in $[0, L_\infty]$ with $L_\infty = b^2(1+WD)^2$.

Uniform deviation. Consider the one-sided suprema $Z^+ = \sup_{\|w\|_2\le W}\big(\widehat{R}_j(w) - R_j(w)\big)$ and $Z^- = \sup_{\|w\|_2\le W}\big(R_j(w) - \widehat{R}_j(w)\big)$ over the validation sample of size m. Changing one sample moves either supremum by at most L_∞/m, so McDiarmid's inequality gives $Z^\pm \le \mathbb{E}Z^\pm + L_\infty\sqrt{\log(4q/\delta)/(2m)}$, each with probability at least $1-\delta/(2q)$. By symmetrization, $\mathbb{E}Z^\pm \le 2\,\mathfrak{R}_m(\mathcal{L})$, the Rademacher complexity of the loss class. The loss is the composition of $t \mapsto t^2$, restricted to $|t| \le b(1+WD)$ where it is $2b(1+WD)$-Lipschitz, with the class $\{u \mapsto w^\top\varphi(u) - Y(u)\}$; translation by the fixed function Y leaves the Rademacher average unchanged (the translated term does not involve w and has zero mean), and for the linear class over the ℓ^2 ball,

$$\mathfrak{R}_m \le \frac{W}{m}\mathbb{E}\Big\|\sum_{i=1}^m \sigma_i\varphi(\tilde{u}_i)\Big\|_2 \le \frac{W}{m}\Big(\sum_{i=1}^m \mathbb{E}\|\varphi(\tilde{u}_i)\|_2^2\Big)^{1/2} \le \frac{WbD}{\sqrt{m}},$$

by Jensen's inequality. Talagrand's contraction lemma then gives $\mathfrak{R}_m(\mathcal{L}) \le 2b(1+WD)\cdot WbD/\sqrt{m}$, hence

$$Z^\pm \le \frac{4Wb^2D(1+WD)}{\sqrt{m}} + L_\infty\sqrt{\frac{\log(4q/\delta)}{2m}} \qquad \text{jointly with probability} \ge 1-\delta/q.$$

Empirical risk minimization. On this event, for any minimizer w^* of R_j over the ball, $R_j(\widehat{w}_j) \le \widehat{R}_j(\widehat{w}_j) + Z^- \le \widehat{R}_j(w^*) + Z^- \le R_j(w^*) + Z^+ + Z^-$, and $Z^+ + Z^-$ is bounded by the displayed excess. A union bound over the two tails of each of the q coordinates makes all events hold simultaneously with probability at least $1-\delta$, and averaging over j preserves the (coordinate-uniform) bound. This proves (i).

For (ii), the ridge minimizer satisfies $\widehat{R}_j(\widehat{w}_j) + \lambda\|\widehat{w}_j\|_2^2 \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2$, so $\widehat{R}_j(\widehat{w}_j) \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2 \le \widehat{R}_j(w^*) + \lambda W^2$. Since by assumption $\|\widehat{w}_j\|_2 \le W$, the uniform event of part (i) applies to $\widehat{w}_j$ and w^* alike, and the chain above goes through with the extra λW^2. □

A.6 Proof of Corollary 6.11

The lower bound is Corollary 6.10. For the upper bound, evaluate the quadratic form at the uniform weights $w_m = 1/M$:

$$\frac{1}{\bar{e}^2}\, w^\top S w = \frac{1}{\bar{e}^2}\Big(\frac{1}{M^2}\sum_m S_{mm} + \frac{1}{M^2}\sum_{m\neq k} S_{mk}\Big) \;\le\; \frac{1+\delta_e}{M} + \frac{M-1}{M}\,\big(\bar{\varrho} + \delta_\varrho\big),$$

using the entrywise upper bounds and the counts M and $M(M-1)$ of the two sums. The right side equals $\bar{\varrho} + \delta_\varrho + \big(1 + \delta_e - \bar{\varrho} - \delta_\varrho\big)/M$, and the minimum over the simplex is at most the value at any feasible point. □

A.7 Proof of Corollary 6.6

Fix a coordinate j and a grid point γ. Substituting the affine form of the correction,

$$h_{w,\gamma}(u)_j = w_j^\top \widetilde{\varphi}_{j,\gamma}(u) + o_{j,\gamma}(u), \qquad \widetilde{\varphi}_{j,\gamma}(u) = \varphi_j(u) - \Phi_j^\top c_\gamma(u), \quad o_{j,\gamma}(u) = c_\gamma(u)^\top Y_j,$$

and neither $\widetilde{\varphi}_{j,\gamma}$ nor $o_{j,\gamma}$ depends on w or on the validation sample: c_γ is built from the training design alone. Each component of $\Phi_j^\top c_\gamma(u)$ is a c_γ-weighted sum of member values bounded by b, so its absolute value is at most $b\,\|c_\gamma(u)\|_1 \le b\kappa$, whence $\|\widetilde{\varphi}_{j,\gamma}(u)\|_2 \le bD + bD\kappa = bD(1+\kappa)$ and $|o_{j,\gamma}(u)| \le b\kappa$. For $\|w\|_2 \le W$ the prediction deviates from the target by at most $WbD(1+\kappa) + b\kappa + b \le b(1+WD)(1+\kappa) =: \tilde{c}$, so the squared losses lie in $[0, \tilde{c}^2]$ and are $2\tilde{c}$-Lipschitz in the prediction.

The argument of Proposition 6.5 now applies verbatim to the class indexed by w in the W-ball, for this fixed (j,γ): the Rademacher complexity of the affine part is at most $W\, bD(1+\kappa)/\sqrt{m}$, contraction multiplies it by $2\tilde{c}$, symmetrization doubles it, and McDiarmid at level $\delta/(2q|\Gamma|)$ adds $\tilde{c}^2\sqrt{\log(4q|\Gamma|/\delta)/(2m)}$. A union bound over the two tails of the $q|\Gamma|$ pairs makes, with probability at least $1-\delta$, each one-sided deviation $Z^{\pm}_{j,\gamma}$, defined as in the proof of Proposition 6.5, at most

$$\frac{4Wb^2D(1+WD)(1+\kappa)^2}{\sqrt{m}} + b^2(1+WD)^2(1+\kappa)^2\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2m}}.$$

On that event, write $\bar{R}(\gamma) = \frac{1}{q}\sum_j R_j(h_{\widehat{w},\gamma})$ and $\widehat{\bar{R}}(\gamma)$ for its empirical version; averaging the $Z_{j,\gamma}$ over j bounds $|\widehat{\bar{R}}(\gamma) - \bar{R}(\gamma)|$ by the same quantity for every γ simultaneously, including at the data-dependent $\widehat{w}$, because the deviations are uniform over the ball. The selection chain $\bar{R}(\widehat{\gamma}) \le \widehat{\bar{R}}(\widehat{\gamma}) + Z \le \widehat{\bar{R}}(\gamma^*) + Z \le \bar{R}(\gamma^*) + 2Z$ with γ^* the aggregate oracle then gives the display, whose constant is $2Z$ in the factored form stated. □

A.8 Proof of Proposition 6.7

Since $\sum_m w_m = 1$, the stack's normalized residual is $\rho_{f_w}(u) = \sum_m w_m \rho_{f_m}(u)$, hence

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 = \mathbb{E}\Big\|\sum_m w_m \rho_{f_m}(u)\Big\|_2^2 = \sum_{m,k} w_m w_k\,\mathbb{E}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle = w^\top S w.$$

For (i), parameterize $w = (1-t, t)$; $q(t) = w^\top S w$ is a convex quadratic in t with $q'(0) = 2(\varrho e_1 e_2 - e_1^2)$, negative precisely when $\varrho < e_1/e_2$. In that case the unconstrained minimizer lies in $(0,1)$ and gives the stated interior value; otherwise $q'(0) \geq 0$, the minimum over $[0,1]$ is at $t = 0$ with value e_1^2, and the second member is dropped. For (ii), $S = e^2\big((1-\varrho)I + \varrho\,\mathbf{1}\mathbf{1}^\top\big)$, so $w^\top S w = e^2\big((1-\varrho)\|w\|_2^2 + \varrho\big)$ on the simplex, minimized by the uniform weights, where $\|w\|_2^2 = 1/M$. □

A.9 Proof of Corollary 6.10

Convex weights are nonnegative, so each term of $w^\top S w = \sum_m w_m^2 S_{mm} + \sum_{m\neq k} w_m w_k S_{mk}$ can be bounded below entrywise:

$$w^\top S w \;\geq\; \bar{e}^{\,2}\sum_m w_m^2 + \bar{\varrho}\,\bar{e}^{\,2}\sum_{m\neq k} w_m w_k = \bar{e}^{\,2}\Big(\|w\|_2^2 + \bar{\varrho}\,(1-\|w\|_2^2)\Big),$$

using $\sum_{m\neq k} w_m w_k = (\sum_m w_m)^2 - \|w\|_2^2 = 1 - \|w\|_2^2$. The bracket is a convex combination of 1 and $\bar{\varrho}$, hence at least $\bar{\varrho}$. □

A.10 Proof of Proposition 6.12

The metric is diagonal, so the squared norm splits over coordinates and the j-th summand $w_j^2\,\mathbb{E}\big(\sum_m v_{mj} f_m(u)_j - G(u)_j\big)^2$ depends on the weights only through the column $v_{\cdot j}$. Minimizing the sum is therefore q independent minimizations, and the positive factor w_j^2 does not move any of the q argmins, so the optimizer is the same for every positive w. A combination with shared weights is the special case $v_{mj} = v_m$ for all j, a subset of the feasible set of each coordinate problem, which gives the domination. □

A.11 Proof of Corollary 6.13

The integrand is nonnegative, so Tonelli's theorem permits exchanging the expectation with the coordinate sum:

$$\mathbb{E}\Big[a(u)\sum_j w_j^2\big(f_v(u)_j - G(u)_j\big)^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\big(f_v(u)_j - G(u)_j\big)^2\Big].$$

The j-th summand depends on v only through its j-th column $v_{\cdot j}$, exactly as in the proof of Proposition 6.12, so minimizing over v splits into q independent problems; the factors w_j^2 rescale the summands without moving any per-coordinate minimizer, which gives the simultaneous optimality over all diagonal w. The choice $a(u) = \|G(u)\|_2^{-2}$, $w = \mathbf{1}$ identifies the objective with $\mathbb{E}\,\|\rho_{f_v}\|_2^2$, the mean squared relative error. Finally $\mathbb{E}\|\rho\|_2 \leq (\mathbb{E}\|\rho\|_2^2)^{1/2}$ is Jensen's inequality. □

A.12 Proof of Proposition 6.14

Fix a coordinate j and a member m. The validation scores $a(\tilde{u}_i)(f_m(\tilde{u}_i)_j - G(\tilde{u}_i)_j)^2$, $i \le m_v$, are i.i.d., take values in $[0, L]$ by assumption, and have mean $R_j(m)$; the members and the weight are fixed relative to the validation sample. Hoeffding's inequality gives

$$\Pr\Big(\big|\widehat{R}_j(m) - R_j(m)\big| \ge t\Big) \ \le\ 2\exp\big(-2m_v t^2/L^2\big)$$

for each of the Mq pairs (m, j). With $t = L\sqrt{\log(2Mq/\delta)/(2m_v)}$ and a union bound, all Mq deviations are below t simultaneously with probability at least $1-\delta$. On that event, for every j, writing $m^*(j)$ for a population minimizer,

$$R_j\big(\widehat{m}(j)\big) \le \widehat{R}_j\big(\widehat{m}(j)\big) + t \le \widehat{R}_j\big(m^*(j)\big) + t \le R_j\big(m^*(j)\big) + 2t.$$

The metric statement follows by multiplying the coordinate-wise inequality by $w_j^2 \ge 0$ and summing; the selection $\widehat{m}(\cdot)$ does not depend on w, so one event serves every diagonal metric. □

A.13 Proof of Proposition 6.16

Let $T : \mathcal{H}_k \to \mathbb{R}^{\mathcal{X}}$ be the composition map $Tf = f \circ \varphi$. For $x \in \mathcal{X}$ the reproducing property gives $(Tf)(x) = f(\varphi(x)) = \langle f, k(\cdot, \varphi(x))\rangle_{\mathcal{H}_k}$, so evaluation of Tf at x is a bounded functional, and the image $\mathcal{H}_{k_\varphi} := T(\mathcal{H}_k)$ carries the quotient norm $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : Tf = g\}$, the minimum over the affine subspace $T^{-1}(g)$ (nonempty exactly when g is a pullback). This normed space is an RKHS: its reproducing kernel is $k_\varphi(x, x') = \langle k(\cdot, \varphi(x)), k(\cdot, \varphi(x'))\rangle_{\mathcal{H}_k} = k(\varphi(x), \varphi(x'))$, since the minimum-norm representer of evaluation at x is $k(\cdot, \varphi(x))$. Taking $f = h$ in the minimum gives $\|h \circ \varphi\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and substituting this bound into Theorem 6.17 applied with the kernel k_φ replaces $\|G\|$ by $\|h\|_{\mathcal{H}_k}$. □

A.14 Proof sketch of Proposition 6.15

This appendix records the argument at the level of the idealization stated with the proposition — projected image marginal bounded above and below, exact rank at the working resolution, and the interpolation limit standing in for the validated nugget — without tracking constants; no lower bound is claimed. The argument has five steps: a pointwise error bound whose design dependence is the power function; a fill-distance bound for random designs; an upper bound on the warped target's native norm; the two substitutions; and the reduction from the validated nugget to the interpolation scale. Throughout, k is a Matérn kernel of smoothness s whose native space on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^{d'}$ is norm-equivalent to $H^{s+d'/2}(\Omega)$ [30, Ch. 10]; we follow the convention of the statement and write s for the resulting approximation order.

Step 1: pointwise bound through the power function. For any target F with components in the native space and the interpolant $\widehat{F}_0$ on the design X, the argument that proves Theorem 6.17 (with $\lambda = 0$ and $m = 0$) gives, for every u,

$$\|F(u) - \widehat{F}_0(u)\|_2 \ \le\ \|F\|_K \, P_0(u),$$

and the classical zeros lemma for functions with many zeros on a dense set converts the power function into the fill distance $h_X = \sup_{u\in\Omega} \min_i \|u - u_i\|$: $\sup_u P_0(u) \le C\, h_X^s$ for h_X below a domain constant [30, Ch. 11]. Hence $\sup_u \|F(u) - \widehat{F}_0(u)\|_2 \le C\, h_X^s\, \|F\|_K$.

Step 2: fill distance of an i.i.d. design. Let $u_1, \ldots, u_n$ be i.i.d. from a density bounded below by $p_- > 0$ on Ω. Fix $\delta > 0$ and a $\delta/2$-net of Ω of cardinality $N(\delta) \le C_\Omega\, \delta^{-d'}$. A net ball of radius $\delta/2$ has μ-mass at least $c\,p_- \delta^{d'}$, so the probability that some net ball receives no sample is at most $C_\Omega\, \delta^{-d'}(1 - c\,p_- \delta^{d'})^n \le C_\Omega\, \delta^{-d'} e^{-c\,p_-\, n\delta^{d'}}$. Choosing $\delta^{d'} = (2/(c\,p_-))\,(\log n)/n$ makes this at most $C'\, n^{-1}$, and on the complement every point of Ω lies within δ of a sample: with probability at least $1 - C'/n$,

$$h_X \;\le\; C'' \Big(\frac{\log n}{n}\Big)^{1/d'}.$$

This is the high-probability statement in the proposition; the logarithm is absorbed into the logarithmic factors there.

Step 3: the warped target's norm, from above. Let $F = g(A\,\cdot)$ with $\|g\|_{H^s} = 1$ (native norm of the isotropic kernel in the image variables). For integer m, the chain rule gives, for any multi-index α with $|\alpha| = m$, $D^\alpha\{g(Au)\} = \sum_{|\beta|=m} c_{\alpha\beta}(A)\,(D^\beta g)(Au)$ with $\sum_\beta |c_{\alpha\beta}(A)| \le \|A\|^m = \sigma_1^m$, so by the change of variables $y = Au$ (Jacobian $|\det A|$),

$$\|D^\alpha\{g(A\cdot)\}\|_{L^2(\mathcal{U})} \;\le\; \sigma_1^m\, |\det A|^{-1/2}\, \|g\|_{H^m(A\mathcal{U})}.$$

Summing over $|\alpha| \le m$ bounds $\|g(A\cdot)\|_{H^m}$ by $C\,\sigma_1^m |\det A|^{-1/2} \|g\|_{H^m}$ for integer m, and the fractional case follows by interpolation between consecutive integers [1, Ch. 7]. Only this upper bound is used; the $|\det A|^{-1/2}$ and the domain constants are among the singular-value constants absorbed into the statement, leaving the displayed σ_1^s.

Step 4: the two designs. For the isotropic kernel on the raw input the domain has dimension $d' = d$, so Steps 1–3 give the first display: $C\,h_X^s\,\sigma_1^s \lesssim \sigma_1^s n^{-s/d}$ up to logarithms. For the adapted kernel $k_A(u, u') = k(Au, Au')$, interpolating F at $\{u_i\}$ is interpolating g at the image design $\{Au_i\}$, and $\|F\|_{K_A} = \|g\|_K \le 1$ by the pullback identity (Proposition 6.16). The image measure has a density bounded above and below on its support by the idealization of the statement. Split $A = A_r + A_\perp$ along the leading-r right singular subspace. The projected points $\Pi_r A u_i$ are i.i.d. on an r-dimensional bounded set, so Step 2 (with $d' = r$) puts every projected image point within $C(\log n/n)^{1/r}$ of a projected sample; and for every u in the domain, $\|Au - \Pi_r Au\| \le \sigma_{r+1}\,\mathrm{diam}(\mathcal{U})$. The triangle inequality then bounds the image fill distance by $C(\log n/n)^{1/r} + 2\sigma_{r+1}\,\mathrm{diam}(\mathcal{U})$, and the approximate-rank hypothesis $\sigma_{r+1} \lesssim n^{-1/r}$ makes the second term of the same order as the first. Steps 1 and 3 with $d' = r$ and $\|g\|_K \le 1$ give the second display. If instead σ_{r+1} is not below $n^{-1/r}$, the trailing directions are resolved by the design, the image fill distance is that of a d-dimensional set, and the two rates coincide; only the constant σ_1^s separates the bounds, as claimed.

Step 5: from the interpolant to the validated ridge estimator. The pipeline fits with a nugget from a grid containing values down to $10^{-8}n$ and selects on validation. For any fixed λ the pathwise bound of Theorem 6.17 controls the ridge estimator by $\|F\|_K\,\sup_u P_\lambda(u)$, and a two-line spectral computation compares the regularized power function with the interpolation one: with $\kappa = k(X, u)$ and $A_\lambda = K + n\lambda I$,

$$P_\lambda(u)^2 - P_0(u)^2 = \kappa^\top\big(K^{-1} - A_\lambda^{-1}\big)\kappa = n\lambda\,\kappa^\top K^{-1} A_\lambda^{-1}\kappa \;\le\; \frac{n\lambda\,\|\kappa\|_2^2}{\lambda_{\min}(K)\,\big(\lambda_{\min}(K) + n\lambda\big)},$$

which vanishes as $\lambda \to 0$. Validation selection over the grid can only improve on its smallest-nugget arm up to the selection cost priced by Lemma 6.4, which is of lower order than the rates displayed. Both displays are upper bounds under the stated idealization, and no matching lower bound is claimed. $\square$

A.15 Proof of Theorem 6.17

Fix u and write $\kappa = k(X,u) \in \mathbb{R}^n$, $A = K + n\lambda I$, $w = A^{-1}\kappa$. For each component j, membership $r_j \in \mathcal{H}_k$ and the reproducing property give

$$r_j(u) - \widehat{r}_{\lambda,j}(u) = \Big\langle r_j,\ k(u,\cdot) - \sum_{i=1}^{n} w_i\, k(u_i,\cdot)\Big\rangle_{\mathcal{H}_k},$$

because $\widehat{r}_{\lambda,j}(u) = \kappa^\top A^{-1} r_j(X) = \sum_i w_i\, r_j(u_i)$ and $r_j(u_i) = \langle r_j, k(u_i,\cdot)\rangle$. Cauchy–Schwarz yields

$$\big|r_j(u) - \widehat{r}_{\lambda,j}(u)\big| \ \le\ \|r_j\|_{\mathcal{H}_k} \Big\|k(u,\cdot) - \sum_i w_i k(u_i,\cdot)\Big\|_{\mathcal{H}_k},$$

and the second factor is independent of j; call it $\rho(u)$. Expanding,

$$\rho(u)^2 = k(u,u) - 2\, w^\top \kappa + w^\top K w = k(u,u) - \kappa^\top A^{-1}\big(2A - K\big)A^{-1}\kappa.$$

Since $2A - K = A + n\lambda I$,

$$\rho(u)^2 = k(u,u) - \kappa^\top A^{-1}\kappa - n\lambda\, \|A^{-1}\kappa\|_2^2 = \widetilde{P}_\lambda(u)^2 \ \le\ P_\lambda(u)^2.$$

Summing the squared componentwise bounds,

$$\big\|r(u) - \widehat{r}_\lambda(u)\big\|_2^2 = \sum_j \big(r_j(u) - \widehat{r}_{\lambda,j}(u)\big)^2 \le \rho(u)^2 \sum_j \|r_j\|_{\mathcal{H}_k}^2 = \widetilde{P}_\lambda(u)^2\, \|r\|_K^2. \quad \square$$

Note that the argument does not use interpolation: it holds for every $\lambda \ge 0$, with the (slightly sharper) factor $\widetilde{P}_\lambda$ showing that smoothing can only tighten this particular bound relative to the posterior standard deviation P_λ that we report.

A.16 Proof of Proposition 6.18

Write $s_i = \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)$ for the validation scores and $s^\star = \|e(u^\star)\|_2 / P_\lambda(u^\star)$ for the test score. Because m, $\widehat{r}_\lambda$, X and P_λ are fixed and the inputs $\tilde{u}_1, \ldots, \tilde{u}_m, u^\star$ are exchangeable and independent of them, the scores $s_1, \ldots, s_m, s^\star$ are exchangeable. The event $G(u^\star) \in C(u^\star)$ is exactly $\{s^\star \le q\}$, where $q = s_{(\lceil (1-\alpha)(m+1)\rceil)}$ is the $k := \lceil (1-\alpha)(m+1)\rceil$-th order statistic of the validation scores.

Consider the augmented sample $s_1, \ldots, s_m, s^\star$ of size $m+1$ and let $\mathrm{rank}(s^\star)$ be its rank (ties broken uniformly at random, which only helps). By exchangeability the rank is uniform on $\{1, \ldots, m+1\}$, so $\mathbb{P}(\mathrm{rank}(s^\star) \le k) = k/(m+1)$. If $s^\star$ is among the k smallest of the $m+1$ values then at most $k-1$ of the s_i are below it, so $s^\star \le s_{(k)} = q$; conversely $s^\star \le q$ forces $\mathrm{rank}(s^\star) \le k$ except possibly through ties. Hence

$$\mathbb{P}(s^\star \le q) \ \ge\ \mathbb{P}(\mathrm{rank}(s^\star) \le k) = \frac{k}{m+1} = \frac{\lceil (1-\alpha)(m+1)\rceil}{m+1} \ \ge\ 1 - \alpha,$$

which is the lower bound. When the scores are almost surely distinct there are no ties, $\{s^\star \le q\} = \{\mathrm{rank}(s^\star) \le k\}$ exactly, and $k/(m+1) < ((1-\alpha)(m+1)+1)/(m+1) = 1 - \alpha + 1/(m+1)$, giving the upper bound. $\square$

A.17 Proof of Corollary 6.21

The nonzero eigenvalues of $K/n = XX^\top/n$ coincide with those of the sample covariance matrix $S = X^\top X/n \in \mathbb{R}^{p\times p}$, and zero eigenvalues contribute nothing to $d_{\mathrm{eff}}(\lambda) = \sum_i \lambda_i(K)/(\lambda_i(K)+n\lambda)$, so

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} = \frac{p}{n}\cdot\frac{1}{p}\sum_{j=1}^{p}\frac{\lambda_j(S)}{\lambda_j(S)+\lambda} = \frac{p}{n}\Big(1-\lambda\,\widehat{m}_S(-\lambda)\Big),$$

by the computation of Lemma 6.20 applied to S, with $\widehat{m}_S$ the Stieltjes transform of the empirical spectral distribution of S. By the Marčenko–Pastur theorem [19], this distribution converges weakly, almost surely, to the law with ratio γ and scale σ^2, and since $x \mapsto (x+\lambda)^{-1}$ is bounded and continuous on $[0,\infty)$, $\widehat{m}_S(-\lambda) \to m(-\lambda)$. It remains to evaluate $m(-\lambda)$. The Stieltjes transform of the Marčenko–Pastur law satisfies the self-consistent equation

$$m(z) = \frac{1}{\sigma^2\big(1-\gamma-\gamma z\, m(z)\big)-z},$$

i.e. $\gamma\sigma^2 z\, m^2 + \big(z-\sigma^2(1-\gamma)\big)m + 1 = 0$. At $z=-\lambda$ the discriminant is $\big(\lambda+\sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda > 0$, and of the two real roots

$$m(-\lambda) = \frac{\sigma^2(1-\gamma)+\lambda \pm \sqrt{\big(\lambda+\sigma^2(1-\gamma)\big)^2+4\gamma\sigma^2\lambda}}{-2\gamma\sigma^2\lambda}$$

only the one with the minus sign in the numerator is positive, as $m(-\lambda) = \int (x+\lambda)^{-1} dF(x) > 0$ requires; rearranging gives the stated expression. □

A.18 Proof of Lemma 6.20

Diagonalize K with eigenvalues λ_i. Then

$$\operatorname{tr}\big(K(K+n\lambda I)^{-1}\big) = \sum_{i=1}^{n}\frac{\lambda_i}{\lambda_i+n\lambda} = \sum_{i=1}^{n}\Big(1-\frac{n\lambda}{\lambda_i+n\lambda}\Big) = n-\lambda\sum_{i=1}^{n}\frac{1}{\lambda_i/n+\lambda} = n\big(1-\lambda\,\widehat{m}(-\lambda)\big),$$

using $\widehat{m}(-\lambda) = \frac{1}{n}\sum_i(\lambda_i/n+\lambda)^{-1}$. The limit statement follows from the weak convergence of the empirical spectral distribution together with the boundedness and continuity of $x \mapsto (x+\lambda)^{-1}$ on $[0,\infty)$ for fixed $\lambda > 0$. □

B Implementation details

The pipeline was developed on a single laptop (one NVIDIA RTX PRO 2000, 8 GB; 16 CPU cores, 64 GB RAM); the structural-mechanics numbers were then recomputed from scratch on CPU-only cloud instances under the seed protocol below, in single precision for network training and double precision for all kernel solves and error computations, and it is those recomputed values that the tables report. One exception is stated where it occurs: the six-member configuration of Table 5 completed at two campaign seeds and is reported from those. The run records do not carry a device field, so the device statements here describe the machines the campaign ran on and are not recoverable from the artifacts. The dataset is the distributed `StructuralMechanics` file pair (40000 samples); inputs are reduced to $\mathbb{R}^{41}$ after verifying the broadcast structure exactly.

Splits: the training block is samples 1–20000, the test set is samples 20001–40000. High-data protocol: a fixed permutation of the training block reserves 1000 samples for validation, 19000 for fitting. Low-data protocol: the first 1250 samples of the training block, of which the last 250 form the validation split. The test set is never touched during development; each configuration is evaluated on it once, after selection on validation.

FNO. Width 64, 14 modes per dimension, 4 spectral layers with pointwise linear skips and GELU, lift from 3 channels (broadcast load, two coordinates), projection $64 \to 128 \to 1$. AdamW, learning rate 1.5×10^{-3} (2×10^{-3} with batch 256), weight decay 10^{-6}, cosine schedule to 10^{-6}, 200 epochs, batch 256. 6.45M parameters.

Transformer (implemented, not in the reported ensemble). Encoder: 41 tokens (load value and 10 sine/cosine pairs of the coordinate), 5 pre-norm blocks, dimension 192, 4 heads, MLP ratio 4. Decoder: grid queries from Fourier features of both coordinates, two cross-attention blocks, pointwise head $192 \to 192 \to 1$; 3.16M parameters. The cross-attention over 1681 grid queries is the most compute-intensive of our means, and the hardware available for this study (a single laptop GPU that proved unstable under sustained load) did not let us train it to the level of the other members; we therefore exclude it from the reported ensemble and note it as the natural fourth architecture family for a follow-up on stable hardware.

UNet. Three convolutional scales ($41 \to 21 \to 11$ by average pooling, ceil mode) and a bottleneck at 6×6; two 3×3 convolutions with GroupNorm(8) and SiLU per block; widths 48/96/192/384; bilinear upsampling with skip concatenation; 1×1 output head. Input channels as for the FNO. AdamW, learning rate 1.5×10^{-3}, weight decay 10^{-5}, cosine schedule, 200 epochs, batch 256. 4.38M parameters.

MSE variant. The residual MLP trained with mean squared error on standardized targets in place of the metric loss, otherwise identical recipe but for the schedule: the campaign ran it for 120 epochs; it enters the ensemble as a loss-diversity member and is also the strongest plain MLP we obtain.

MLP. Input layer $41 \to 1024$, three residual SiLU blocks of width 1024, output $1024 \to 1681$. AdamW, learning rate 10^{-3}, weight decay 10^{-5}, cosine schedule, 400 epochs, batch 256. 4.91M parameters. A wider variant (width 1536, five residual blocks, 14.5M parameters) is trained under the same recipe.

Refiner. The same residual trunk with input layer $(41+1681) \to 1024$: the load is concatenated with the kernel method's stress prediction for that load. Training uses four-fold out-of-fold kernel predictions for the input channel and full-data kernel predictions at evaluation; otherwise identical to the MLP but run for 100 epochs (the trainer's own default of 300 is not what the campaign used). 6.64M parameters. Reflection augmentation flips the load and permutes both the kernel channel and the target by the row-reversal index.

Common training details. Loss: mean over the batch of $\|\hat{v} - v\|_2 / \|v\|_2$ in original units (predictions denormalized inside the loss; targets standardized by the per-pixel training mean and global standard deviation), except for the MSE variant above. Reflection augmentation with probability 1/2; reflection averaging at evaluation. Model selection: validation error computed every 10 epochs, best checkpoint kept. The ensemble's diversity comes from the architectures and losses rather than from reseeding; the correlation analysis of Section 4.6 indicates same-architecture reseeds would be more correlated still.

Seed protocol. One seed value enters every stochastic component at once: the permutation that draws the validation split from the training pool, each member's initialization and batch order, the half-split of the stacking comparison, the subsamples used to tune the kernel stages, and the conformal calibration draw; the pool/test boundary itself never moves. ClimSim is the one exception, and deliberately so: that benchmark fixes no test split of its own, so the seed there also draws the 20000-row test block out of the full set (`campaign/climsim_seeded.py`). Its seed spreads therefore include test-sampling variation, which the structural-mechanics and OCO-2 spreads exclude by construction; that makes them the more conservative of the two conventions where ClimSim is concerned, and it is why the ClimSim seed scatter is quoted as a check on the crossing rather than as a precision claim. Tables report the mean and standard deviation across seeds. What the repository carries is the per-seed result record for every stage of every run — the JSON files the tables and this appendix are computed from, together with the scripts that compute them. The prediction arrays and network checkpoints those records were derived from are too large to distribute and are not included, so a reader can recheck every reported number against the records and can rerun the pipeline from the code, but cannot rescore a new rule on our stored predictions without retraining. Where a statement below turns on an array rather than on a record, it is marked.

What the campaign completed. The structural-mechanics chain was enqueued at ten seeds on four CPU instances and the members completed at all ten: the kernel ridge, the metric-loss MLP, the normalized-MSE MLP, and the refiner, forty trained artifacts in total. The campaign's 36-hour per-task wall clock then landed inside the FNO stage at every seed, which is why the two remaining members of the published configuration are in different states. The FNO has a best-validation checkpoint at all ten seeds, written by the trainer each time validation improved; those checkpoints are finalized here by running the tail of the trainer on them (evaluate, then save the named artifact), so the FNO is a ten-seed member and nothing about it is retrained. The UNet was scheduled after the FNO, and the wall clock reached it at only two of the ten seeds; it is the one member of Table 5's six-member row that the campaign supplies at two seeds rather than ten, and completing that row would require training eight more UNets. The FNO's own training curves are worth recording because they bound what the truncation cost: its best validation epoch falls between 10 and 70 of the scheduled 200, and at every seed validation was already rising again by the last epoch reached, in several cases sharply (seed 3, 0.0489 at its best against 0.0550 at epoch 150; seed 5, 0.0483 against 0.0557). The member is therefore the early-stopped optimum a full run would also have kept, with the caveat that the cosine schedule never annealed and a late anneal might have found more. With the FNO finalized, all ten seeds carry the five-member stack, the residual correction, the second-moment measurement and the conformal calibration; the four-member versions of each are retained and reported so that the FNO's contribution is a measurement rather than an assumption. Two tasks were lost to conditions worth recording because they shape what a reader should expect from a rerun: one instance was killed by the operating system while forming the 19000×19000 Gram matrix through full-size temporaries (the released code fills it in row blocks instead, Section 4.10), and four seeds hit the wall clock during stack stages and were completed on a larger instance afterwards. The OCO-2 and ClimSim sweeps of Section 5.3 report their own seed counts, which are ten per band and five, three and three by training size respectively. The low-data chain completed at ten seeds. The OCO-2 head pipeline is replicated at ten seeds on each of the three bands — thirty band-seeds at a matched 250-epoch budget, with one further run at 750 epochs to separate budget effects from ordering effects. Each band-seed writes its member predictions and per-sample errors, which is what makes the combination rules, the floors and the scoreboard of Section 5 recomputable without retraining; those arrays stay with the run and are not part of the distributed release. Seven of the thirty band-seeds were additionally run a second time, on different hardware, as a queue was being drained. That accident is a useful control, and we report it as measured: six of the ten rows reproduce to the printed precision across the two runs, including the kernel rows, the flat-metric network, its feature head and the combination that the paper reports, while the three weighted-loss members and their heads move by up to 0.39 points. Seeding fixes the initialization and the data order but not every reduction the training performs, so a seeded rerun of those members should be expected to land nearby rather than exactly; the tables' standard deviations, which are over seeds, are larger than this in every case.

Reconciling the stored arrays. Every member's saved prediction arrays are checked against the metric recorded in its own training record before any ensemble is formed: the array is rescored under the project's metric and the two must agree. The check matters because each seed has its own validation partition, and scoring one seed's predictions against another's split produces errors near 28% that are otherwise easy to mistake for a training failure — and, worse, produces no discrepancy at all on the one seed whose split happens to be the reference. Reconciliation is run at every seed and every member. On the test block, which no seed moves, all forty pairs agree with their recorded reflection-averaged error to between 10^{-14} and 6×10^{-8}, the precision at which the arrays are stored.

Stacking. Global weights: convex, initialized at the simplex minimizer of the measured second-moment matrix S (Proposition 6.7) and polished by a short random search on the validation metric. Per-pixel weights: affine in the members at each grid point, ridge parameter 10^{-3}, fitted on half the validation split and accepted only if they beat the global weights on the held-out half, then refitted on the full split.

Kernel stages. Matérn-5/2 on standardized inputs. The residual correction searches the scale grid $\{1, 2, 4\}\times$ the median pairwise distance (estimated on 2000 points) and the nugget grid $\{10^{-6}, 10^{-5}, 10^{-3}\}$ (scaled by n); the low-data chain of Table 3 uses $\{0.5, 1, 2, 4\}$ with nuggets $\{10^{-7}, 10^{-5}, 10^{-3}\}$. Both are tuned on validation using an 8000-sample subsample of the training set, refit at the chosen pair on the full training set in double precision. The pure kernel baseline (Table 2) uses the grid $\{0.5, 0.75, 1, 1.5, 2\}\times$median and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$ directly at $n = 19000$. Feature stage: penultimate activations of the ensemble members (FNO: spatially averaged pre-projection channels, 128; transformer: mean-pooled encoder tokens, 192; MLP: trunk output, 1024), concatenated, standardized, same kernel family and tuning.

Uncertainty. Posterior standard deviation (2) computed from the Cholesky factor of $K + n\lambda I$ by triangular solves. Conformal calibration uses a random 1000-sample subset of the test block that no training, checkpointing, stacking, or tuning step ever touched; the exact split-conformal quantile of the scores on that subset sets the band, and coverage is evaluated on the disjoint remainder. The hypotheses of Proposition 6.18 therefore hold as stated, with no approximate-exchangeability caveat.

C Numerical verification of the stated results

Every result stated in Section 6 is checked numerically, and the checks are released with the code. They come in two kinds. The first kind builds a synthetic object whose ground truth is known exactly and tests the stated inequality or identity on it: the certified bound of Theorem 6.17 is evaluated on a residual constructed inside the native space with exactly known norm, at several nuggets, where any pointwise violation beyond numerical precision would falsify the statement; the leave-half-out identity and unbiasedness of Proposition 6.2 are checked by direct enumeration; the effective-dimension identity of Lemma 6.20 is checked to machine precision and the closed form of Corollary 6.21 against the empirical spectrum of an i.i.d.-entry Gram matrix; and the rate mechanism of Proposition 6.15 is exercised on a controlled warped target $G = g(A\,\cdot)$ with a known singular-value profile, where the adapted metric must dominate at every sample size and improve the empirical rate. All four pass: the largest relative violation of the optimal-recovery bound is -3.3×10^{-2}, that is, the bound holds with margin everywhere it was evaluated; the kernel-flow identity closes to 2.8×10^{-14}; the effective-dimension identity to 3.7×10^{-15} and its Marčenko–Pastur closed form to 4.4×10^{-5} across five nuggets spanning four orders of magnitude; and the adapted metric dominates the isotropic one at every sample size of the warped construction.

The second kind evaluates the statements whose subject is the pipeline itself on the pipeline's own released artifacts, at every seed the campaign completed. The symmetrization inequality of Proposition 6.1 is checked on every trained member. It is a statement about the expected loss, so a finite test sample may go the other way, and at ten seeds one does: over the thirty members that carry a reflection, averaging lowers test error in twenty-nine cases, by 0.16 points on the metric-trained network, 0.07 on the FNO and 0.02 on the normalized-MSE one, and raises it in the thirtieth by 0.001 points. That single reversal is the sampling noise the proposition permits and a single-seed check cannot see. The second-moment identity of Proposition 6.7 is measured on the validation residual matrix at each seed and its predicted risk compared with the realized risk of the weights it implies; the ratio is 0.938 ± 0.003 over ten seeds. The simplex minimization behind that comparison is solved by sequential quadratic programming under an explicit sum-to-one constraint. The routine released with the earlier version of this work used projected gradient with clipping and renormalization, which is not a projection onto the simplex; on the five-member matrix it terminated 0.8–3.6% above the true minimum, at near-vertex points, and the weight vectors it reported were wrong in a way that read as a finding rather than as a numerical failure. The check that caught it is the cheapest one available and is now part of the suite: evaluate the objective at the returned point, at every pure-member vertex, and at a properly solved optimum, and require the first to be no worse than the others. The bracket of Corollaries 6.10 and 6.11 is evaluated against the realized convex stack at each seed, and the two-member rule of Proposition 6.7(i) is scored prospectively on every member pair of every seed rather than on selected examples — decided on the validation split and checked on the test block, 840 pairs over the thirty OCO-2 band-seeds. Section 5.2 reports what that scoring found, which is not a uniform verdict: the rule's accuracy is governed by the margin between the two quantities it compares. The selection conditions of Propositions 6.12, 6.14 and 6.5 are checked against the realized selections, with the measured constants b, W and κ of Corollary 6.6 and the certificate evaluated next to the realized excess; κ is a supremum over a test probe and therefore a measured lower bound, 31.2 ± 0.9 over the three seeds at which it was evaluated at the working configuration (29.3 at a single-seed probe run outside the campaign) and falling to 3.0 at the largest nugget and length scale of the grid. The coverage of Proposition 6.18 is computed at every seed and each observed value placed against the exact Beta law of Remark 6.19: at the 90% level ten of ten fall inside the central band $[88.1\%, 91.8\%]$, and at the 95% level ten of ten inside $[93.6\%, 96.3\%]$. Finally the data-scaling sweep of the two raw-input kernel rows is refit at nested training sizes and at ten seeds per band. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, a rank ratio between 0.54 and 0.60, so Proposition 6.15 would permit a rate gain here and not merely a constant; the sweep finds the constant and not the rate, and Section 5.1 reports it that way.

The bound comparisons are reported as measured constants, bound value and realized excess side by side; where an oracle risk is needed it is estimated by a deterministic refit on the test arrays, which can only make the reported ratios conservative. The synthetic checks report pass or fail against explicit tolerances; the artifact checks report the measured quantities themselves, together with the number of seeds behind each.